



Drug information

By: Dr. Riffat Ibrahim Riffat



Where can the pharmacist work?

A pharmacist can work in many places such as:

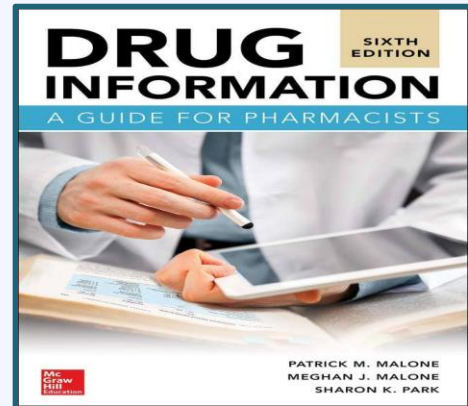
- Community Pharmacy
- Clinical Pharmacy
- Hospital Pharmacy
- Industrial pharmacy
- Academic Pharmacy
- Research and development
- Herbal Pharmacy .
- Drug information centers





Drug information center (DIC)

- A service unit that is operated by trained healthcare professionals who are committed to providing drug information as it relates to therapies, education ,pharmacoeconomics and research programs.



Drug information center (DIC)

- History:

- First DIC was developed in University of Kentucky in 1960.
- In United states 80% of the Hospitals having DIC.
- Egyptian drug information center (EDIC) establishment was organized in 2012



Importance of DIC:

- Access to medical information by health care team is facing some problems:
 - Getting information from textbooks is time consuming and textbooks themselves are expensive and may be outdated
 - Most free websites are not credible
 - Information about drugs from pharmaceutical companies maybe biased.

- thus the demand for rapid, credible and accurate drug information increased.
- WHO supports independent DIC as a core component of national programs to promote better healthcare to all.

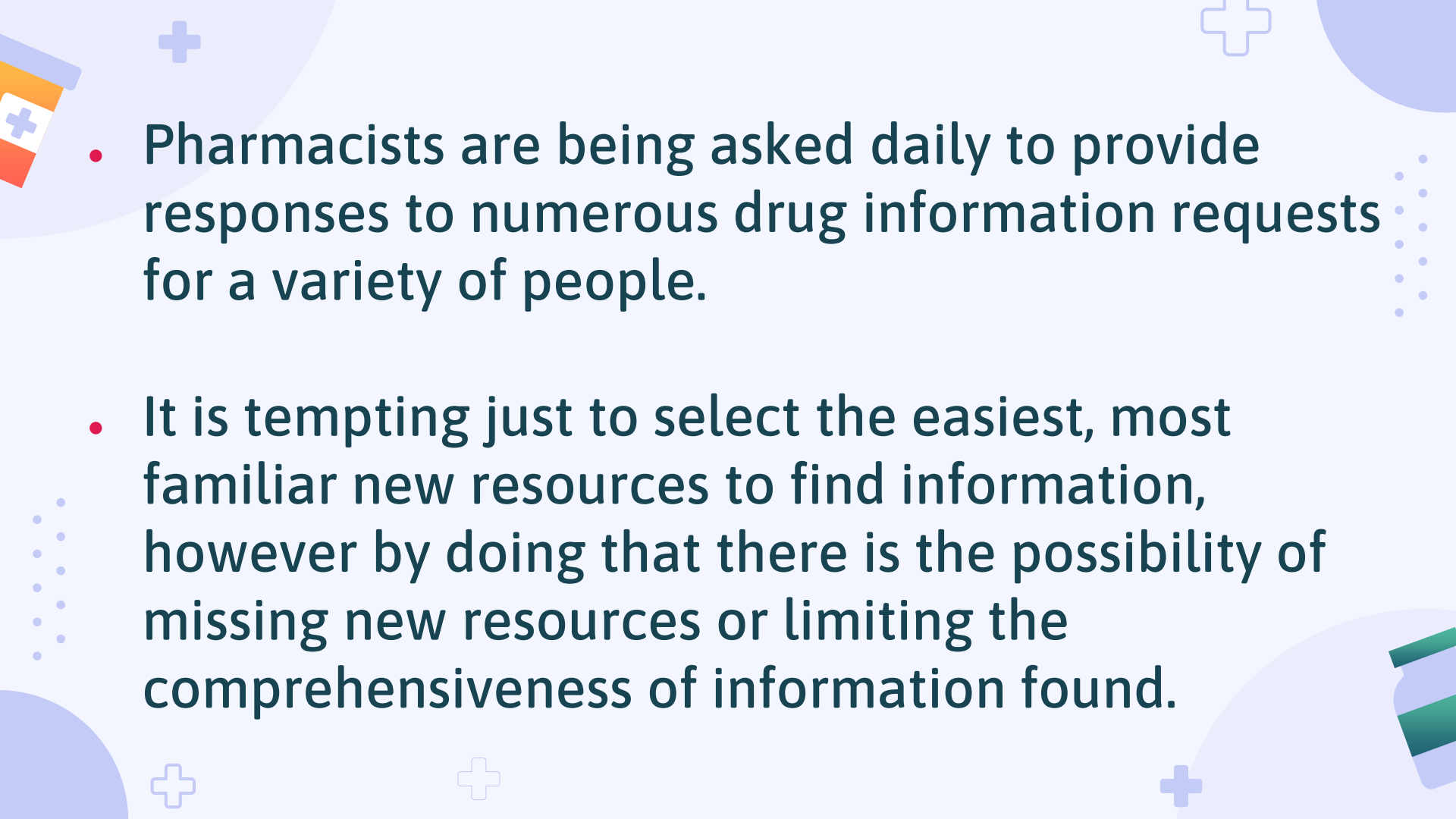


Requirements for DIC

- Team
- Place and hardware
- Methods of communication
- Sources: The information should be properly updated and maintained.



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- The efficient use of drug information (DI) is an important skill for all pharmacists to have regardless of their practice site.
 - In all pharmacy settings, pharmacists are recognized as drug experts and as providers of DI. It is imperative that pharmacists know how to provide accurate and complete responses to DI requests.
 - Keeping current with DI resources is challenging for the clinician because of the vast amount and the variable quality of available resources.
 - Technology has also brought DI to the patient's bedside. Pharmacists should know what DI resources are available and be able to use these sources effectively and efficiently.
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- Pharmacists are being asked daily to provide responses to numerous drug information requests for a variety of people.
 - It is tempting just to select the easiest, most familiar new resources to find information, however by doing that there is the possibility of missing new resources or limiting the comprehensiveness of information found.

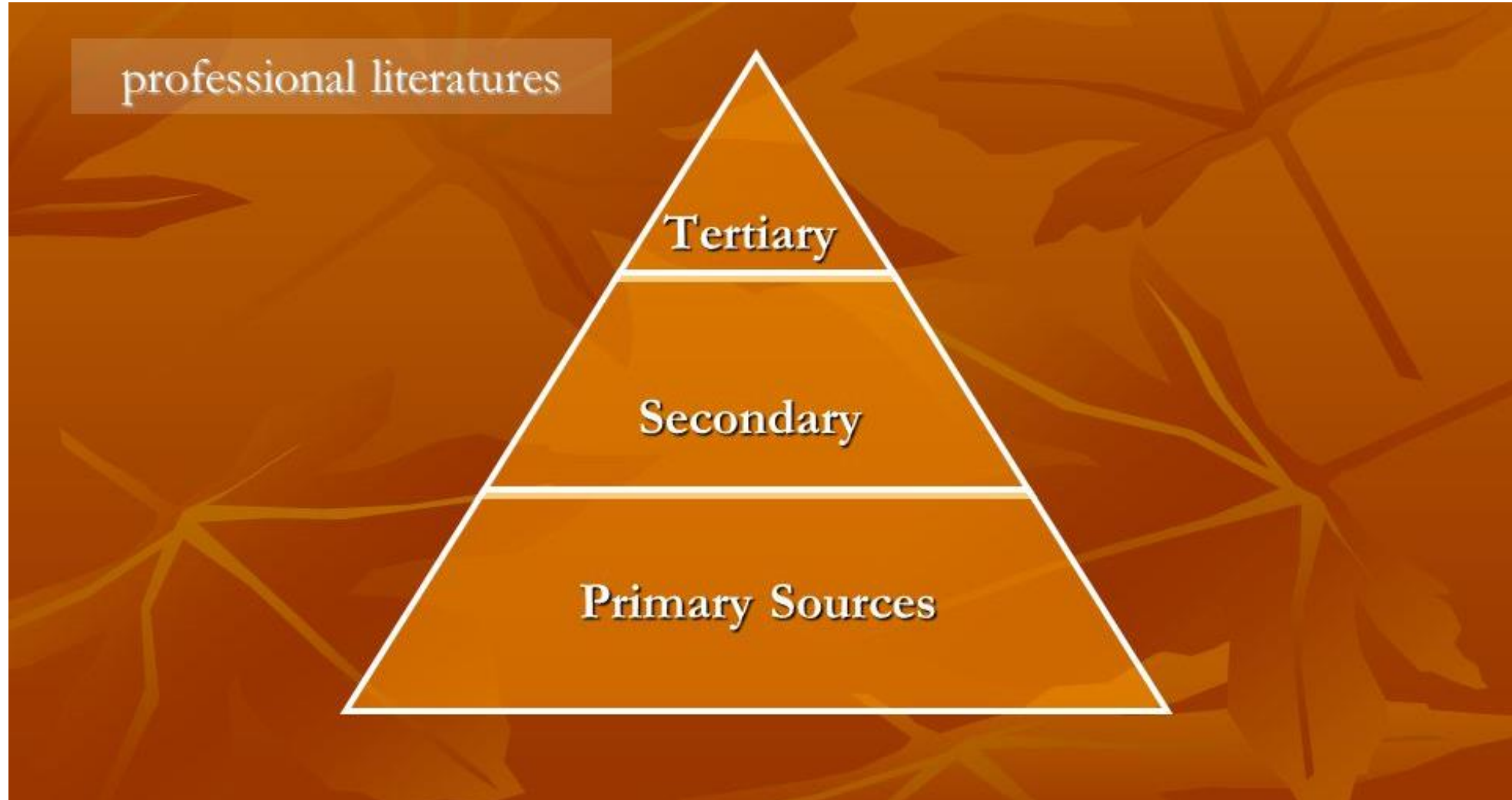
We can **define drug information** as a knowledge of facts through reading, study, or practice experience on chemical substance that is used in diagnosis, prevention and treatment of a disease.

- **Drug Information Service:**

It describes the activities undertaken by pharmacists in providing information to optimize drug use. It is those services which cover the activities of specially trained individual to provide accurate, unbiased, factual information in response to patient oriented drug problems that are received from health care professionals.

the best method to find information includes a stepwise approach moving first through **tertiary** (e.g. textbooks, full-text databases, and review articles) then **secondary** (e.g. indexing or abstracting service) and finally **primary** (e.g. clinical studies) literatures.

Drug information Sources :



Tertiary resources :

-provide information that has been summarized and distilled by the author or editor to provide a quick easy summary of a topic These include not only textbooks and compendia but also electronic databases (e.g., Micromedex, Lexicomp) and review articles.

-The best tertiary resources are written by experts in the field and are peer reviewed. If the tertiary resource is not current or comprehensive, a secondary resource should be consulted to locate primary literature on the topic. However, some questions can only be answered by using tertiary sources.

Advantages of Tertiary Resources:

These references may often serve as an initial place to identify information, due to:

1. They provide a fairly complete and concise overview of information available on a specific topic.
2. These resources are also convenient.
3. easy to use
- 4.familiar to most practitioners.

Disadvantages of Tertiary Resources:

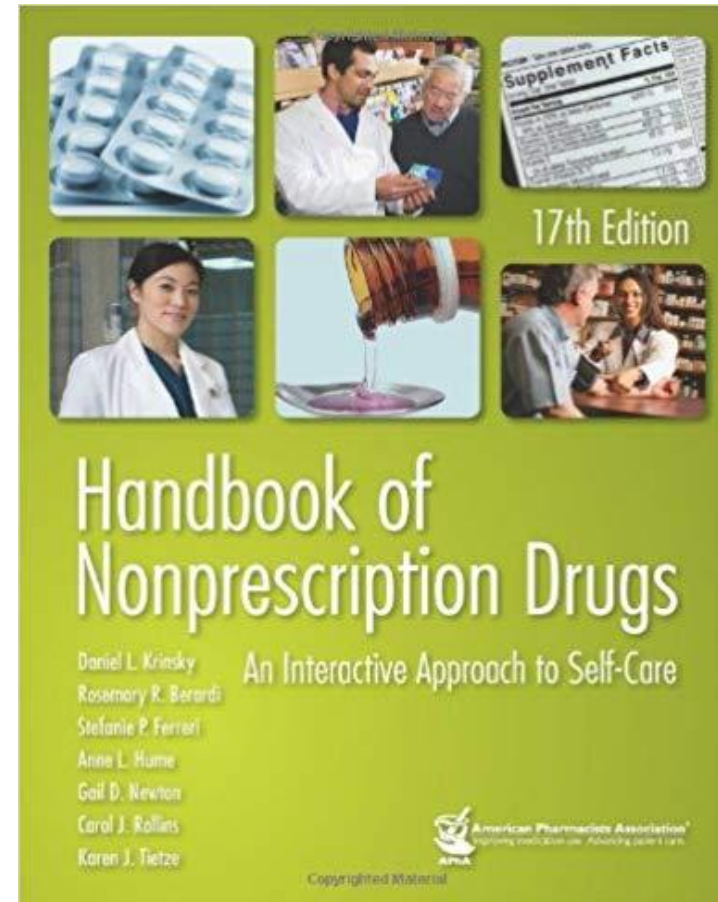
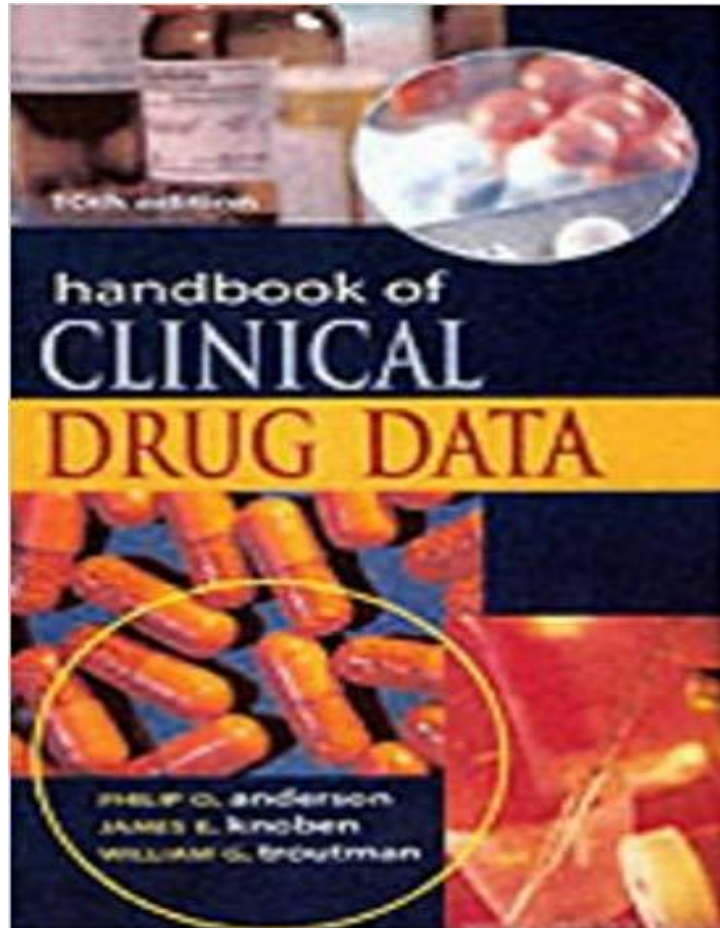
1. The lag time associated with publication of print copy tertiary resources, resulting in less current information.
2. The potential for author biases to be introduced
3. The fact that the information may be incomplete, misleading, or inaccurate .

Evaluation of Tertiary Literature

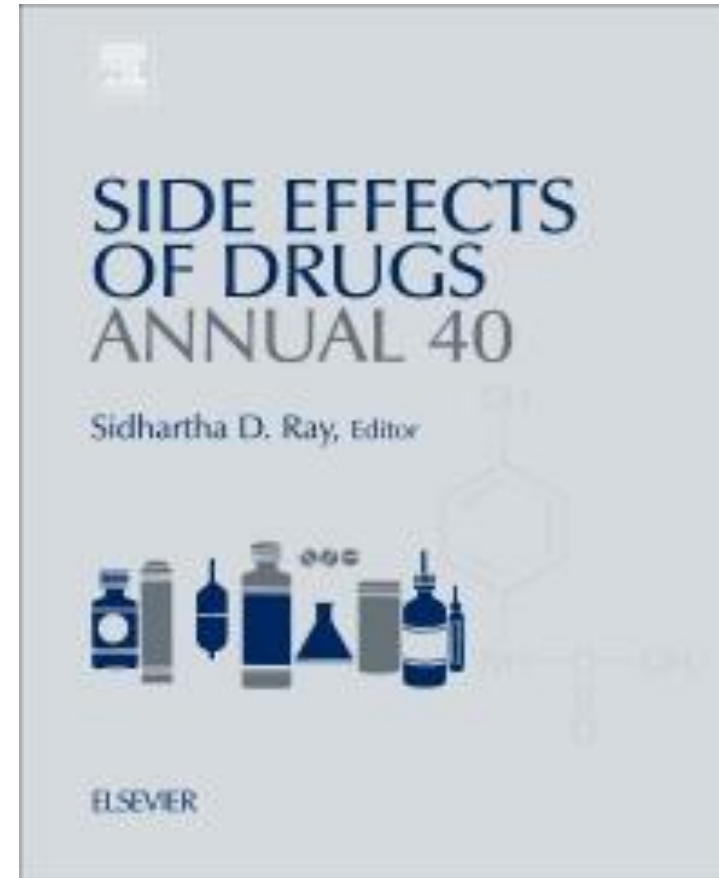
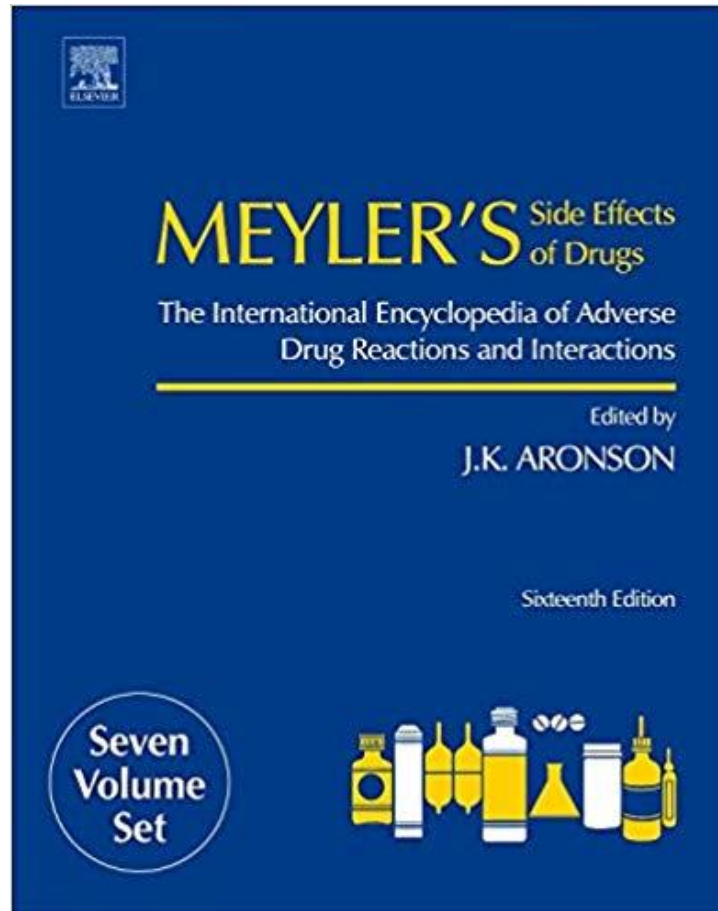
- 1- Does the author have appropriate •
experience/expertise to publish in this area? •
- 2- Is the information likely to be timely based on publication date?
- 3- Is the information supported with •
citations? •
- 4- Does the resource contain relevant information? •
- 5- Is the resource free of bias or errors? •
- 6- Is it clear/concise/easy-to-use ? •

*Example of tertiary drug sources :

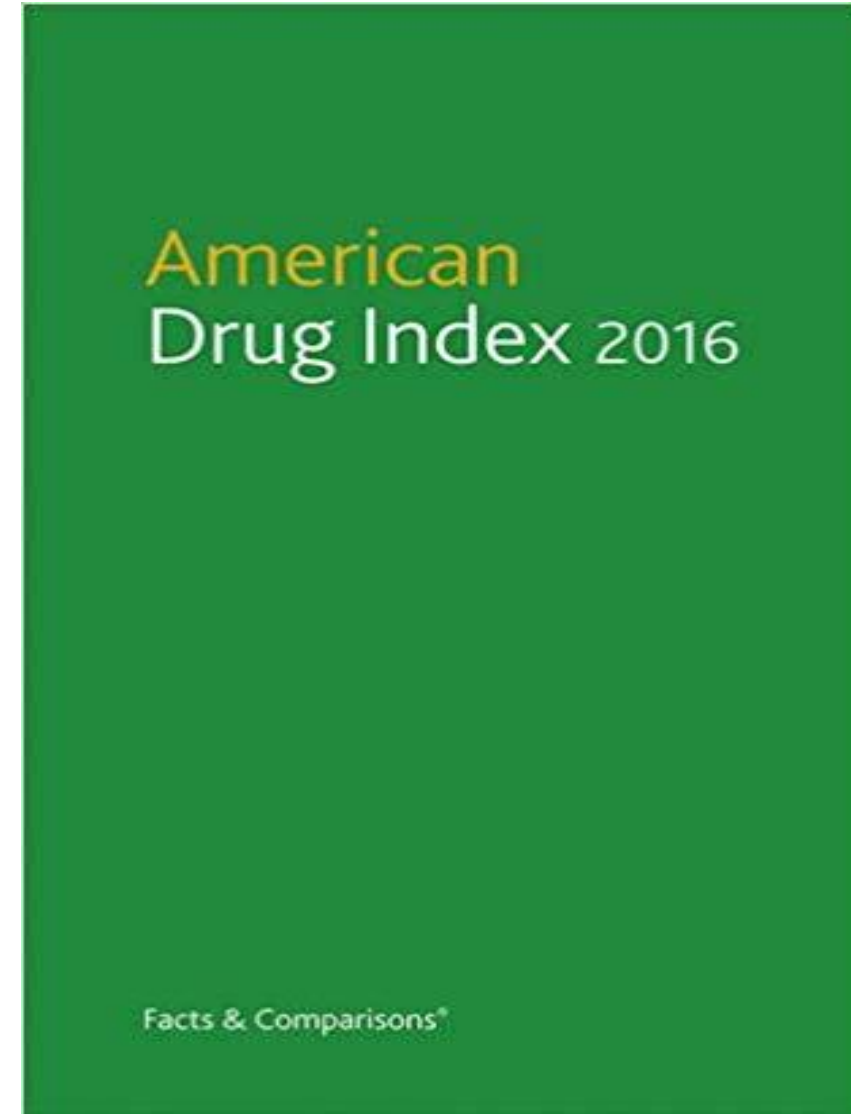
1-General product information



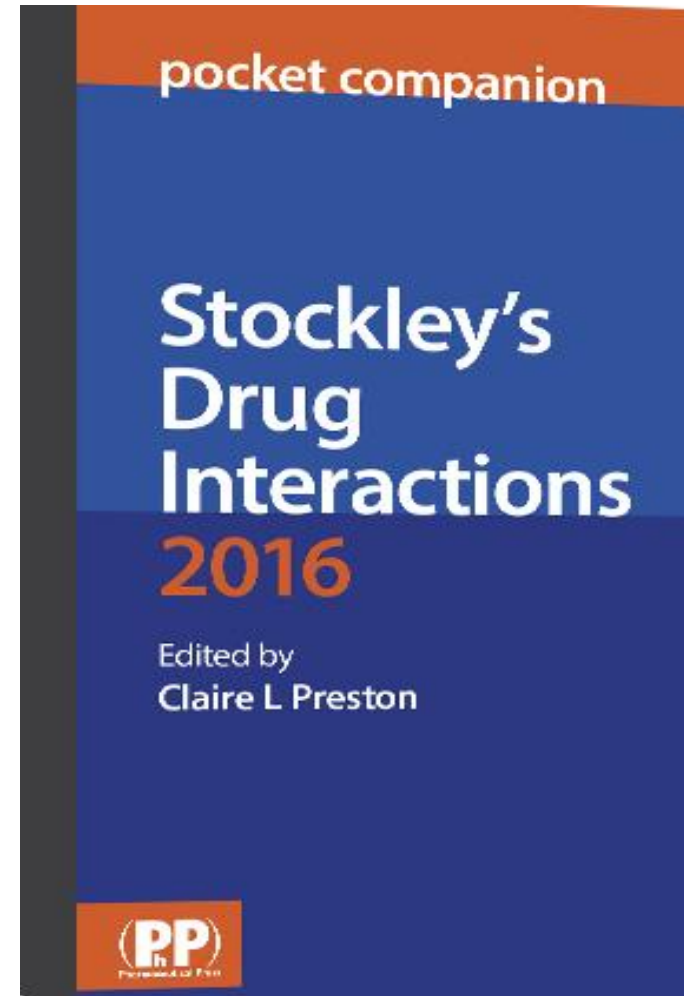
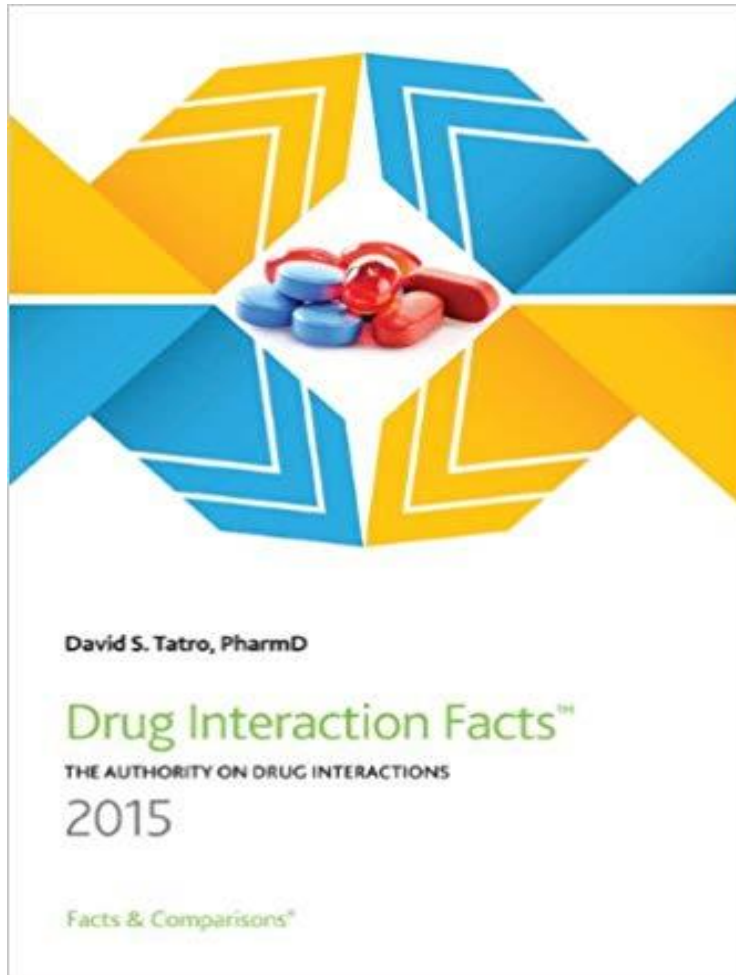
2-Adverse effects



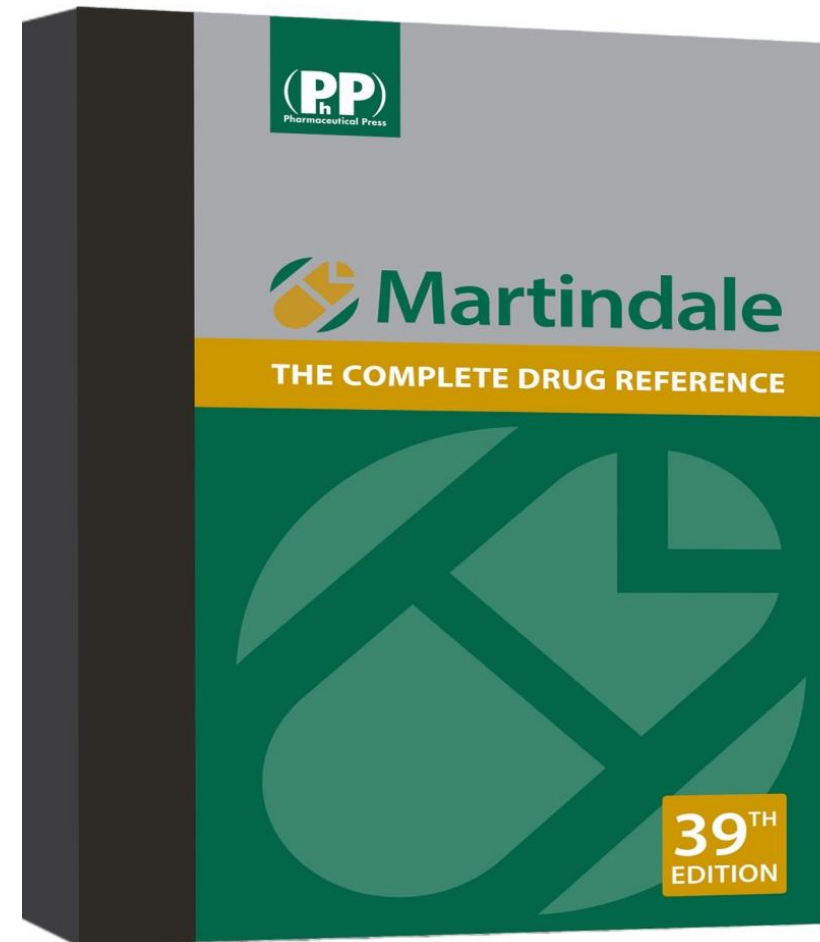
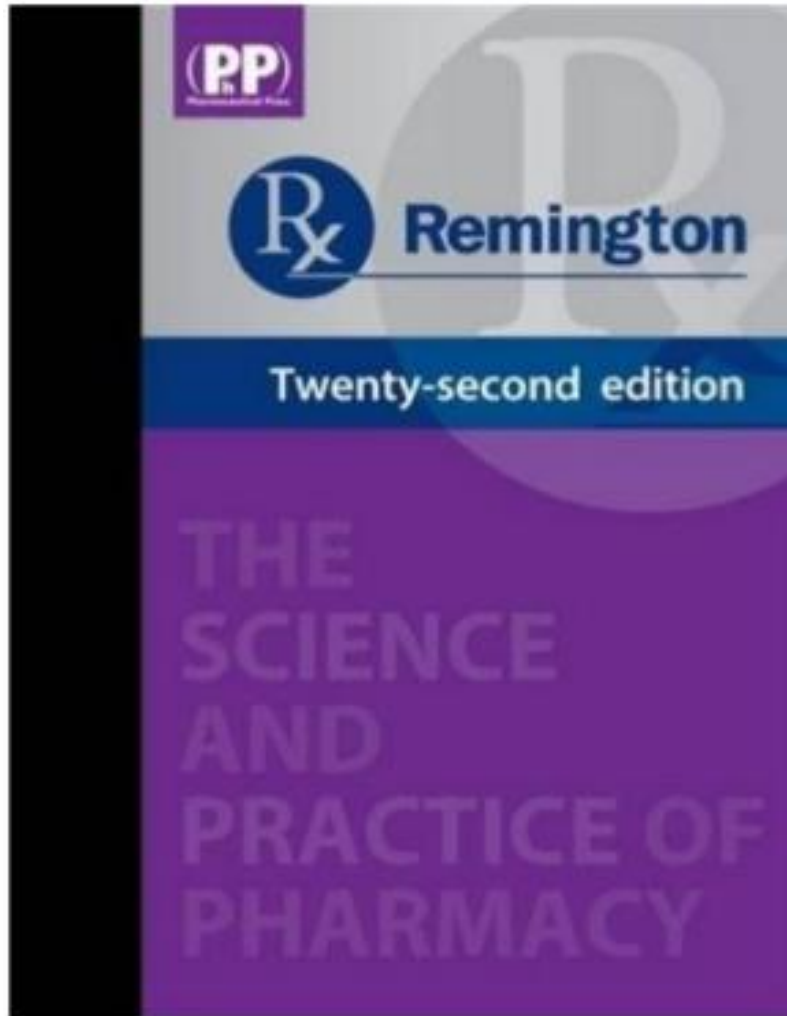
3-Availability of dosage form



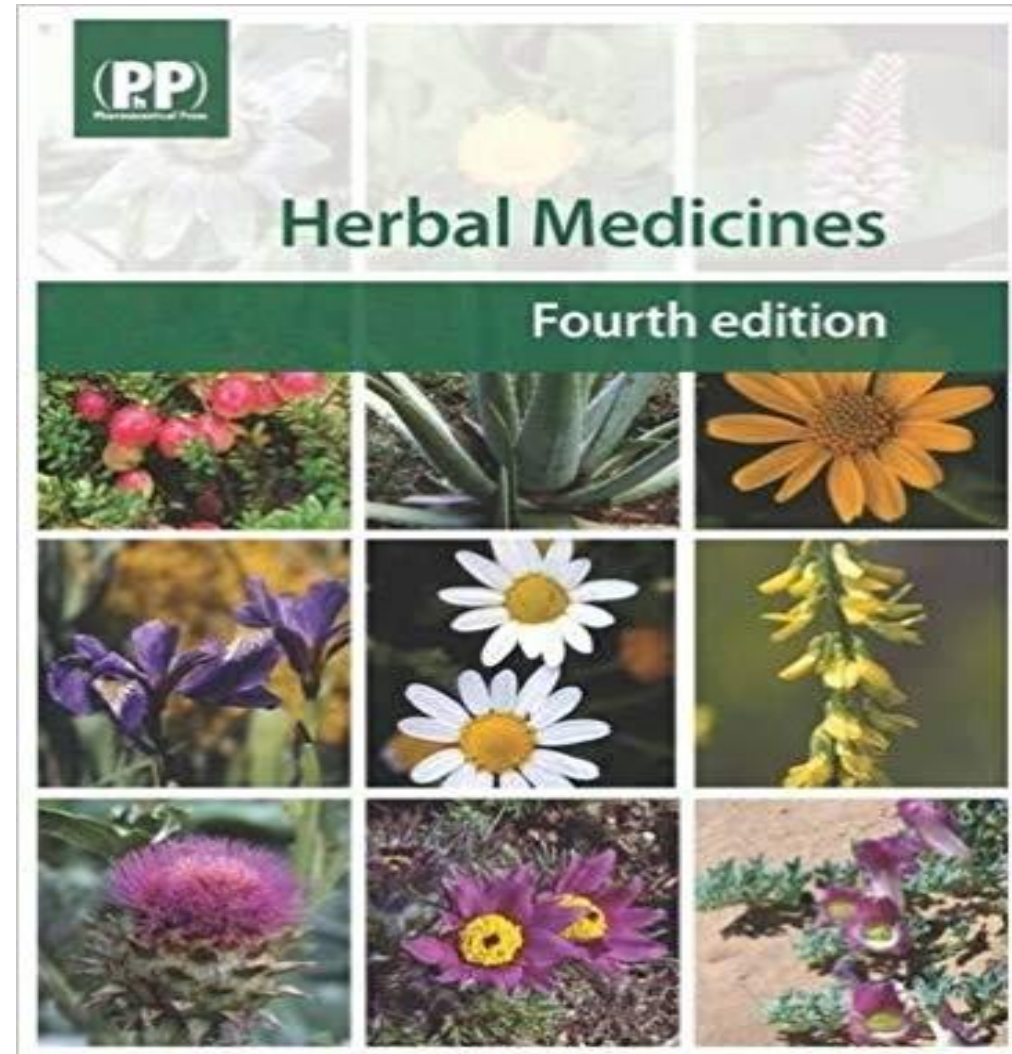
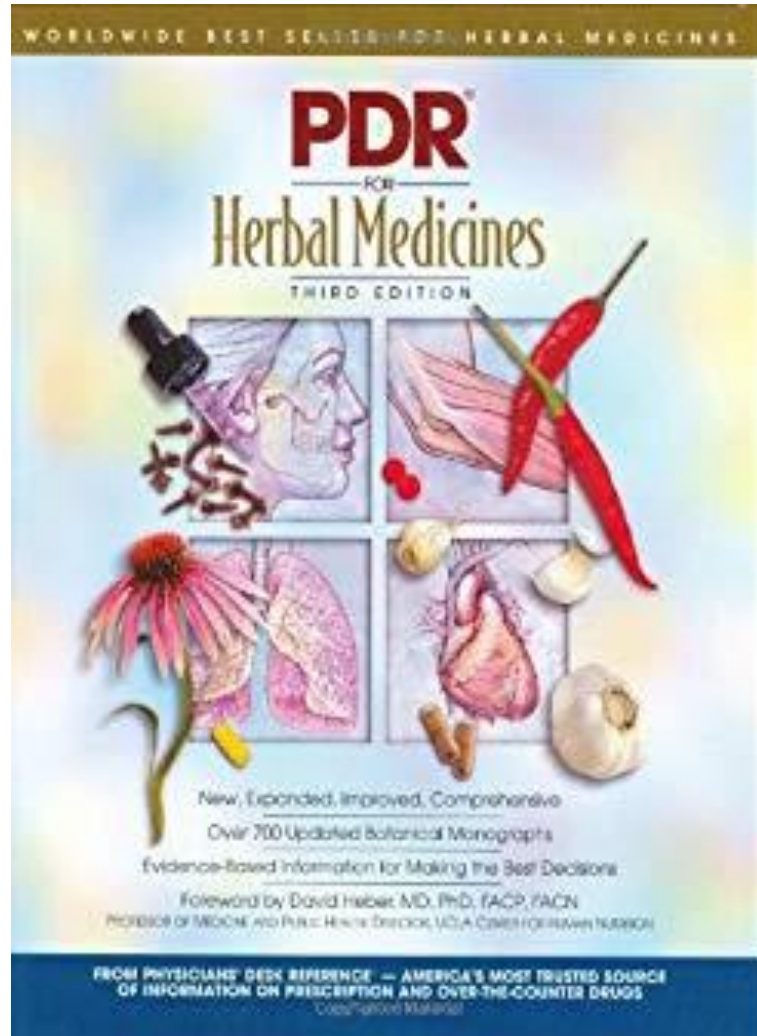
4-Drug interactions



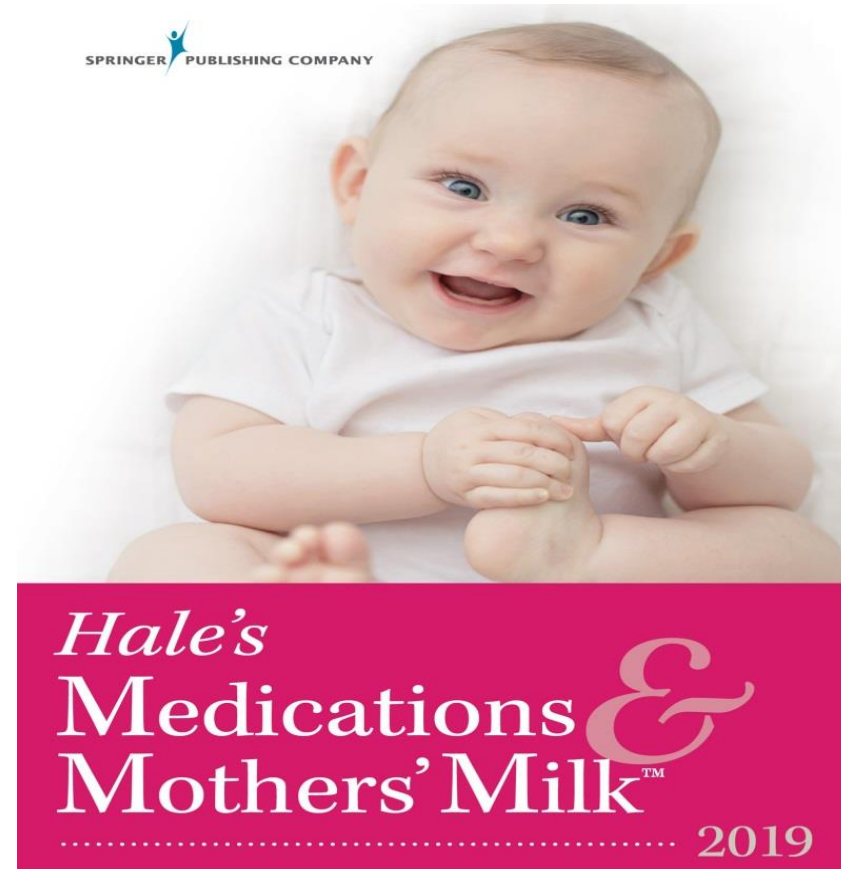
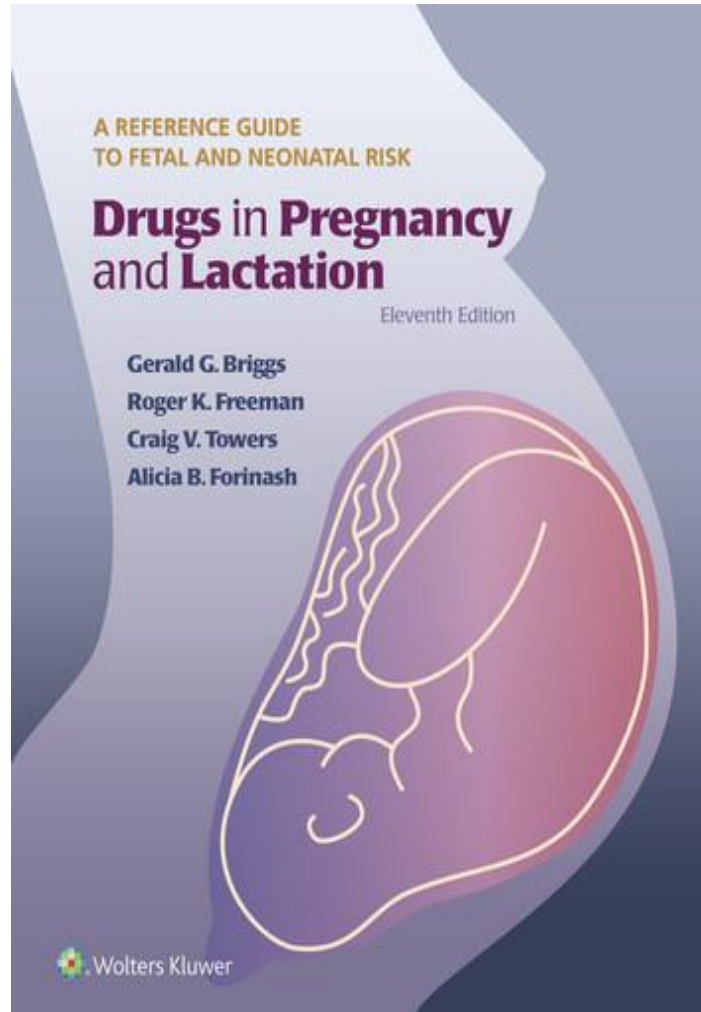
5-Formulations



6-Dietary supplements

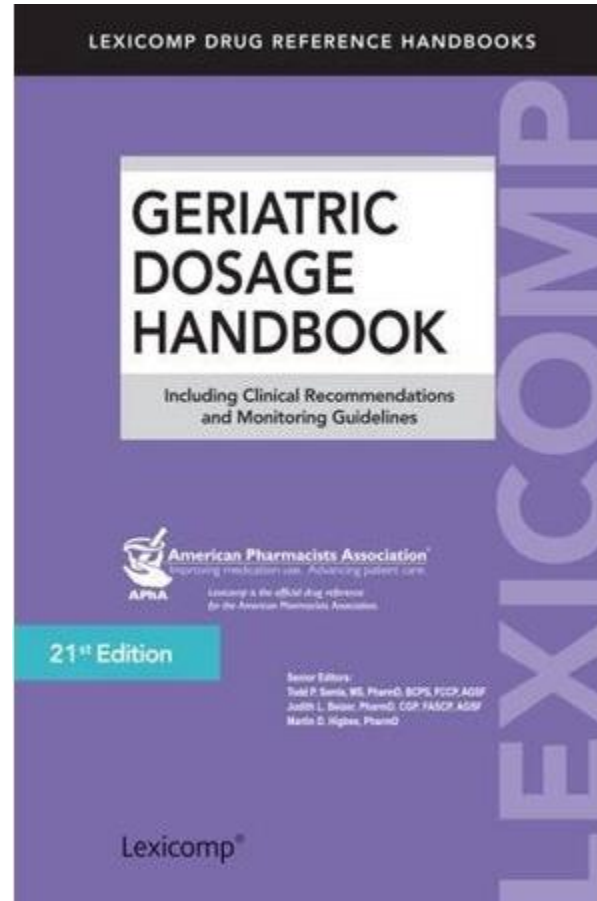


7-Drugs in pregnancy & lactation

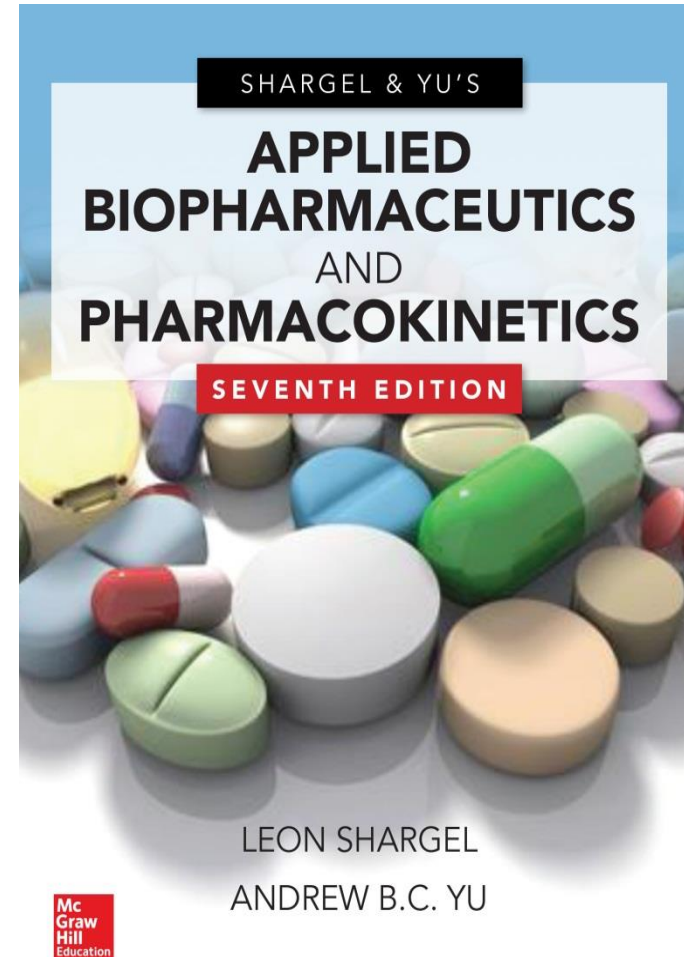
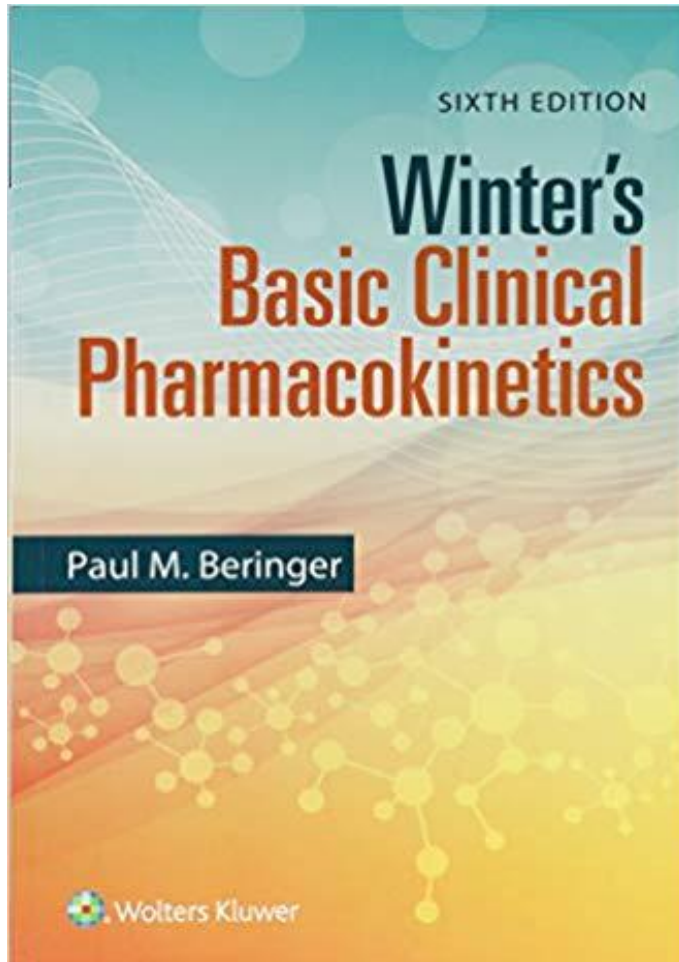


Thomas W. Hale, R.Ph., Ph.D.

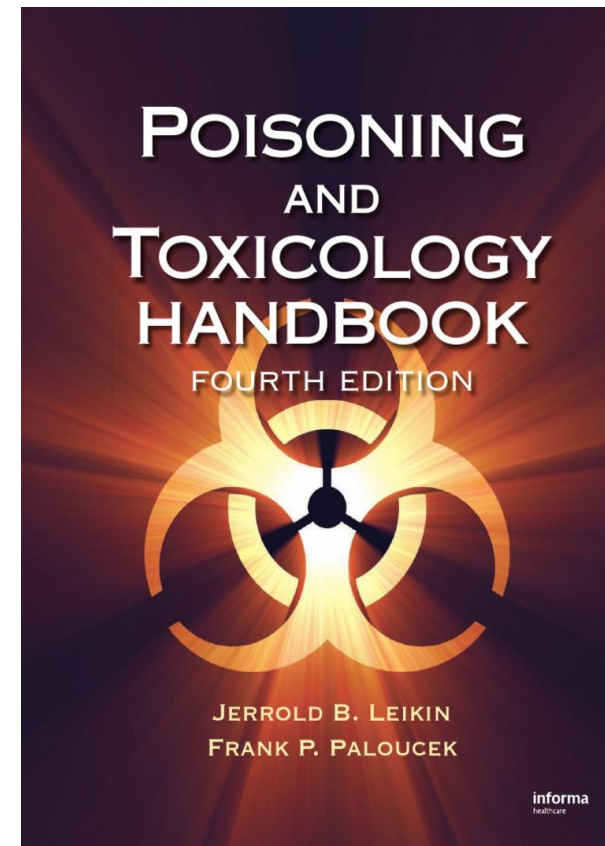
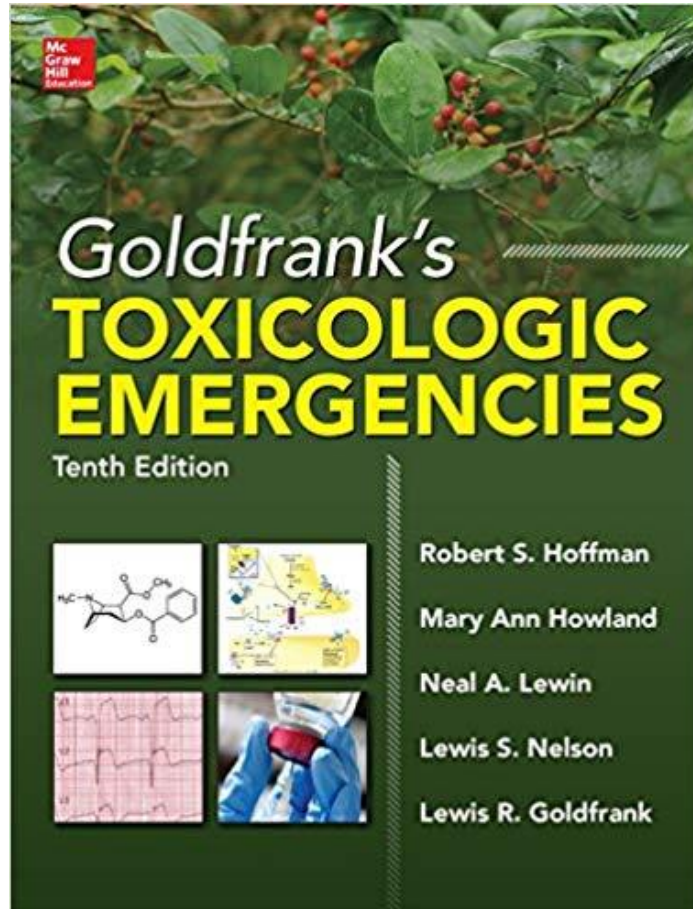
8-Geriatric dosage recommendations



9-Pharmacokinetics



10-Toxicology information



Electronic and Online tertiary information resources

Advantages of online tertiary information resources

- 1.They may be easier to use
- 2. Allow quicker access to information
- 3. Allow multiple searches to be performed simultaneously
- 4. Often contain the most recent information available regarding a topic

Medscape®

Medscape Registration

https://www.medscape.com/features/mktg/public/register/ppc1?src=ppc_bing

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Activate Windows
Go to Settings to activate Windows.

TOXNET from NIH

The screenshot shows the TOXNET website interface. At the top, there is a navigation bar with the NIH logo, the text "U.S. National Library of Medicine", and the "TOXNET TOXICOLOGY DATA NETWORK" logo. To the right of the logo are links for "Mobile", "Help", "FAQs", "TOXNET Fact Sheet", and "Training Manual & Schedule". Below the navigation bar is a large banner area with a molecular structure image on the left and the text "Welcome to TOXNET" on the right, followed by the subtitle "Your resource for searching databases on toxicology, hazardous chemicals, environmental health, and toxic releases". A "Share" button is located in the top right corner of the banner. Below the banner is a search section with three tabs: "SEARCH TOXNET" (selected), "BROWSE", and "ADVANCED SEARCH". The "SEARCH TOXNET" tab contains a search input field with the placeholder text "e.g. benzene, endocrine disruptor", a dropdown menu labeled "ALL DATABASES", and a "Search" button. To the right of the search section is a sidebar with a section titled "Environmental Health & Toxicology" featuring a photo of a child and a "Visit Site" button. Below the search section is a section titled "TOXNET Databases" with two sub-sections: "MOST VISITED BY TOXNET USERS" and "HSDDB" (Hazardous Substances Data Bank). The "HSDDB" section describes it as a peer-reviewed toxicology data bank for over 5,000 hazardous chemicals. Below this is the "TOXLINE" section, which describes it as a database of 4 million references to literature on biochemical, pharmacological, physiological, and toxicological effects of drugs and other chemicals. At the bottom right, there is a "Did you know" section with a question mark icon and text stating: "There is a guide to choosing a database. Which Resource Should I Use can help you pick the right resource for your search." A watermark "Activate Windows" is visible in the bottom right corner of the browser window.

TOXNET

U.S. National Library of Medicine

TOXNET TOXICOLOGY DATA NETWORK

Mobile | Help | FAQs | TOXNET Fact Sheet | Training Manual & Schedule

Welcome to TOXNET

Your resource for searching databases on toxicology, hazardous chemicals, environmental health, and toxic releases

Share

SEARCH TOXNET Search all or select specific databases

BROWSE

ADVANCED SEARCH

e.g. benzene, endocrine disruptor

ALL DATABASES

Search

Environmental Health & Toxicology

Resources on environmental health and toxicology

Visit Site

TOXNET Databases

MOST VISITED BY TOXNET USERS

HSDDB

Hazardous Substances Data Bank. Peer-reviewed toxicology data for over 5,000 hazardous chemicals

TOXLINE

4 million references to literature on biochemical, pharmacological, physiological, and toxicological effects of drugs and other chemicals

Did you know

There is a guide to choosing a database

Which Resource Should I Use

can help you pick the right resource for your search.

Activate Windows

Go to Settings to activate Windows.

<http://www.nlm.nih.gov/>

Medline from NIH

The screenshot shows the MedlinePlus website in a web browser. The browser's address bar displays <https://medlineplus.gov/>. The website's header features the NIH logo and the text "U.S. National Library of Medicine". Below this is the MedlinePlus logo, which includes a stylized green and blue icon and the text "MedlinePlus® Trusted Health Information for You". A search bar with the placeholder text "Search MedlinePlus" and a green "GO" button is located to the right of the logo. Below the search bar are links for "About MedlinePlus", "Site Map", "FAQs", and "Customer Support". A navigation bar contains links for "Health Topics", "Drugs & Supplements", "Videos & Tools", and a language option "Español". The main content area is divided into several sections. On the left, there are three green boxes with white icons: an apple for "Health Topics" (Find information on health, wellness, disorders and conditions), a pill for "Drugs & Supplements" (Learn about prescription drugs, over-the-counter medicines, herbs, and supplements), and a play button for "Videos & Tools" (Discover tutorials, health and surgery videos, games, and quizzes). Below these is a section for "Lab Test Information" with a flask icon (Learn why your doctor orders laboratory tests and what the results may mean). On the right, there is a "Share MedlinePlus" section with icons for email, Facebook, and Twitter. Below this is a featured article titled "Check out our Creamy Squash Soup with Shredded Apples recipe." with a photo of a bowl of soup and text describing it as a quick-to-fix soup bursting with warm-you-up-flavor. Below the article are five small numbered boxes (1, 2, 3, 4, 5) and a play button icon. Further down, there is a "Tweets by @MedlinePlus" section showing a tweet from MedlinePlus.gov about Acute Flaccid Myelitis (AFM). To the right of the tweets is a "Stay Connected" section with a sign-up form for the "My MedlinePlus newsletter" and a green "GO" button. At the bottom right, there is a "NIH MedlinePlus" logo and a small image of a person.

MedlinePlus - Health In

<https://medlineplus.gov/>

NIH U.S. National Library of Medicine

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Health Topics
Find information on health, wellness, disorders and conditions

Drugs & Supplements
Learn about prescription drugs, over-the-counter medicines, herbs, and supplements

Videos & Tools
Discover tutorials, health and surgery videos, games, and quizzes

Lab Test Information
Learn why your doctor orders laboratory tests and what the results may mean

Share MedlinePlus

Check out our **Creamy Squash Soup with Shredded Apples** recipe.
This quick-to-fix soup is bursting with warm-you-up-flavor.

1 2 3 4 5

Tweets by @MedlinePlus

MedlinePlus.gov
@medlineplus

Acute Flaccid Myelitis (AFM) is in the news with the CDC investigating an increase of cases. What symptoms should you be on the look out for? [ow.ly/N6bn30mh0vA](https://www.cdc.gov/media/releases/2014/s091104-afm.html)

Stay Connected
Sign up for the My MedlinePlus newsletter

Enter email address GO

NIH MedlinePlus

FDA website

The screenshot shows a web browser window with multiple tabs. The active tab is titled "Information by Drug Class" and shows the URL <https://www.fda.gov/Drugs/DrugSafety/InformationbyDrugClass/ucm165107.htm>. The browser's address bar also displays this URL. The page content is from the U.S. Food & Drug Administration (FDA) website, which is part of the U.S. Department of Health and Human Services. The header includes the FDA logo, the text "U.S. FOOD & DRUG ADMINISTRATION", and links for "A to Z Index", "Follow FDA", and "En Español". A search bar labeled "Search FDA" is also present. Below the header is a navigation menu with links to Home, Food, Drugs, Medical Devices, Radiation-Emitting Products, Vaccines, Blood & Biologics, Animal & Veterinary, Cosmetics, and Tobacco Products. The main content area is titled "Drugs" and shows a breadcrumb trail: Home > Drugs > Drug Safety and Availability > Information by Drug Class. A blue button labeled "Information by Drug Class" is visible. The main heading is "Acetaminophen Information". Below this heading are social media sharing buttons for Facebook (SHARE), Twitter (TWEET), LinkedIn (LINKEDIN), Pinterest (PIN IT), Email (EMAIL), and Print (PRINT). The text describes Acetaminophen as an active ingredient in hundreds of over-the-counter (OTC) and prescription medicines, used to relieve pain and fever. It also mentions that in prescription medicines, it is often combined with other active ingredients to treat moderate to severe pain, and that the FDA has taken action to improve the safety of consumers when using acetaminophen. On the left side, under the heading "Resources for You", there are four links: "Questions and Answers: FDA warns of rare but serious skin reactions with the pain reliever/fever reducer acetaminophen", "Consumer Education: Using Acetaminophen and Nonsteroidal Anti-inflammatory Drugs Safely", "Sometimes Drugs and the Liver Don't Mix", and "A Guide to Safe Use of Pain Medicine". At the bottom, there is a section titled "Pain Medicine Use During Pregnancy" with a link to "FDA Drug Safety Communication: FDA has reviewed possible risks of pain medicine use during pregnancy 1/9/2015". The browser's status bar at the bottom shows the URL <https://www.fda.gov/Radiation-EmittingProducts/default.htm>.

U.S. Department of Health and Human Services

FDA U.S. FOOD & DRUG ADMINISTRATION

A to Z Index | Follow FDA | En Español

Search FDA

Home Food Drugs Medical Devices Radiation-Emitting Products Vaccines, Blood & Biologics Animal & Veterinary Cosmetics Tobacco Products

Drugs

Home > Drugs > Drug Safety and Availability > Information by Drug Class

Information by Drug Class

Acetaminophen Information

SHARE TWEET LINKEDIN PIN IT EMAIL PRINT

Acetaminophen is an active ingredient in hundreds of over-the-counter (OTC) and prescription medicines. It relieves pain and fever. And, it is also combined with other active ingredients in medicines that treat allergy, cough, colds, flu, and sleeplessness. In prescription medicines, acetaminophen is found with other active ingredients to treat moderate to severe pain. Acetaminophen can cause serious liver damage if more than directed is used. The FDA has taken action to improve the safety of consumers when using acetaminophen.

Resources for You

- Questions and Answers: FDA warns of rare but serious skin reactions with the pain reliever/fever reducer acetaminophen
- Consumer Education: Using Acetaminophen and Nonsteroidal Anti-inflammatory Drugs Safely
- Sometimes Drugs and the Liver Don't Mix
- A Guide to Safe Use of Pain Medicine

Pain Medicine Use During Pregnancy

- FDA Drug Safety Communication: FDA has reviewed possible risks of pain medicine use during pregnancy 1/9/2015

Activate Windows
Go to Settings to activate Windows.

Daily Med (side effects)

The screenshot shows a web browser window with three tabs: "FDA Drugs", "DailyMed", and "FDA Orange Book: Approved Dr". The address bar displays "https://dailymed.nlm.nih.gov/dailymed/". The website header includes the NIH logo and "U.S. NATIONAL LIBRARY OF MEDICINE", along with links for "REPORT ADVERSE EVENTS" and "RECALLS". The main content area features the "DAILYMED" logo and a navigation bar with "ALL DRUGS", "HUMAN DRUGS", and "ANIMAL DRUGS". A large search bar prompts users to "Enter drug, NDC code, drug class, or Set ID". Below the search bar, a section titled "MORE WAYS TO SEARCH:" includes links for "ADVANCED SEARCH", "BROWSE DRUG CLASSES", "LABEL ARCHIVES", and "TABLET/CAPSULE ID TOOL". A message states: "This website contains 105556 drug listings as submitted to the Food and Drug Administration (FDA). At the present time, this Web site does not contain a complete listing of labels for approved prescription drugs." The page is divided into two columns. The left column, titled "NEWS", features a link to "DailyMed Announcements" and a post dated "December 19, 2017" about "Drug Listing Certification". The right column, titled "FDA GUIDANCES & INFORMATION", includes links for "Drug Guidance, Compliance & Regulatory Information" and "View FDA Structured Product Labeling Resources". A section titled "NLM SPL RESOURCES" is also visible at the bottom.

NIH U.S. NATIONAL LIBRARY OF MEDICINE

REPORT ADVERSE EVENTS | RECALLS

DAILYMED

ALL DRUGS | HUMAN DRUGS | ANIMAL DRUGS

Enter drug, NDC code, drug class, or Set ID

MORE WAYS TO SEARCH: ADVANCED SEARCH BROWSE DRUG CLASSES LABEL ARCHIVES TABLET/CAPSULE ID TOOL

This website contains 105556 drug listings as submitted to the Food and Drug Administration (FDA).
At the present time, this Web site does not contain a complete listing of labels for approved prescription drugs.

SHARE

NEWS

[DailyMed Announcements](#)

Posted: December 19, 2017

Drug Listing Certification

The U.S. Food and Drug Administration is reminding the pharmaceutical industry of the December 31, 2017, deadline to update or certify their drug listings with FDA. This applies to drug listings that were not initially listed or updated during the current calendar year. This is the first deadline of the annual certification requirement under

FDA GUIDANCES & INFORMATION

[Drug Guidance, Compliance & Regulatory Information](#)

[View FDA Structured Product Labeling Resources](#)

[View FDA Drug Labeling Guidances](#)

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NLM SPL RESOURCES

The following Structured Product Labeling (SPL) resources have

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 **Pill Identifier Tool** >
Quick, Easy, Pill Identification

 **Drug Interaction Tool** >
Check Potential Drug Interactions

 **Pharmacy Locator Tool** >
Including 24 Hour, Pharmacies

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Secondary resources

- include indexing and abstracting systems that organize and provide easy retrieval of primary resources.
- Indexing systems include the article citation, with or without access to the abstract; some include a link to the full-text article.
- Abstracting systems provide not only the citation but also the abstract and often a link to the full-text article.

Advantages:

- Simple search strategies
- Very current citation information
- Access point for tremendous amount of primary sources

Disadvantages:

- Often require more expertise to use than primary or tertiary resources.
- Retrieved references must be filtered for quality.
- Pharmacists should obtain and evaluate the original article because abstracts may not include enough information.
- Too complex for patients

PubMed Central

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PubMed Central (PMC) is an archive of articles that have been deposited in this free repository. Over 4.8 million articles are available. Access is via journal A-Z listing and keyword searches for articles are entered into the PMC search box.

Other Resources
[PMC International](#)
[Text Mining Collections](#)
[Developer Resources](#)
[NLM LitArch](#)
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are archived in PMC.
Content provided in part by:

2110 <i>Full Participation Journals</i>	330 <i>NIH Portfolio Journals</i>	4623 <i>Selective Deposits</i>
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☒ Articles (☒ include patents) ☐ Case law

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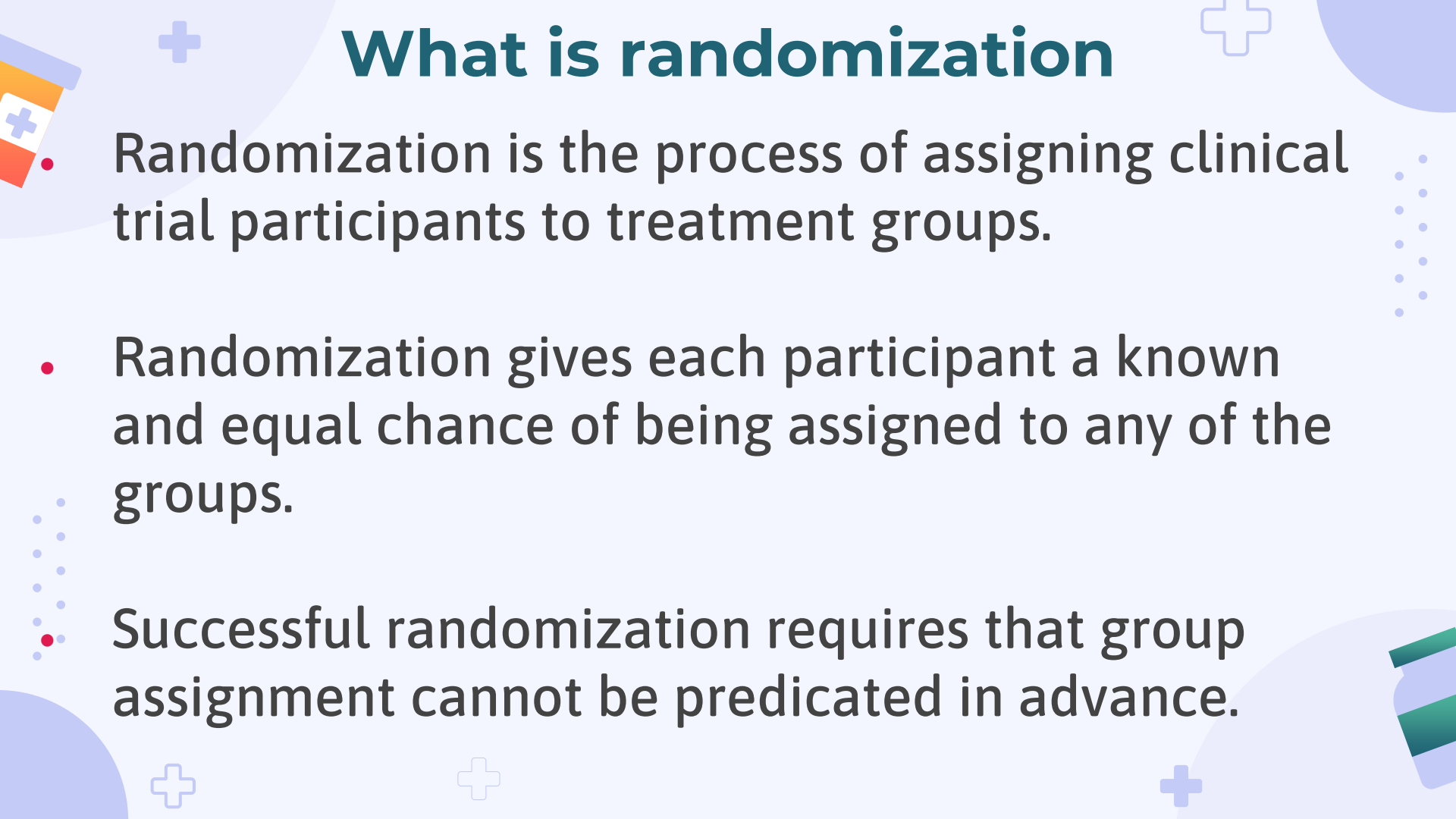
Activate Windows

Go to PC settings to activate Windows.



Drug information

By: Dr. Riffat Ibrahim Riffat



What is randomization

- Randomization is the process of assigning clinical trial participants to treatment groups.
- Randomization gives each participant a known and equal chance of being assigned to any of the groups.
- Successful randomization requires that group assignment cannot be predicated in advance.

Advantages of randomization

- Elimination of selection Bias.
- Allows for comparability between groups.
- Balances arms with respect to prognostic values.

Disadvantages of randomization

- Trials which test for efficacy may not be widely applicable.
- Trials which test for effectiveness are larger and more expensive.
- Results may not always mimic real life treatment situation (e.g. inclusion / exclusion criteria; highly controlled setting)



Methods of randomization



- Simple method randomization.
- Unequal allocation randomization.
- Stratified randomization.
- Blocked randomization.



Simple Randomization

Just what it sounds like:

- Flip a coin
- Random number tables

Typically done before study starts

- Sequence determined in advance

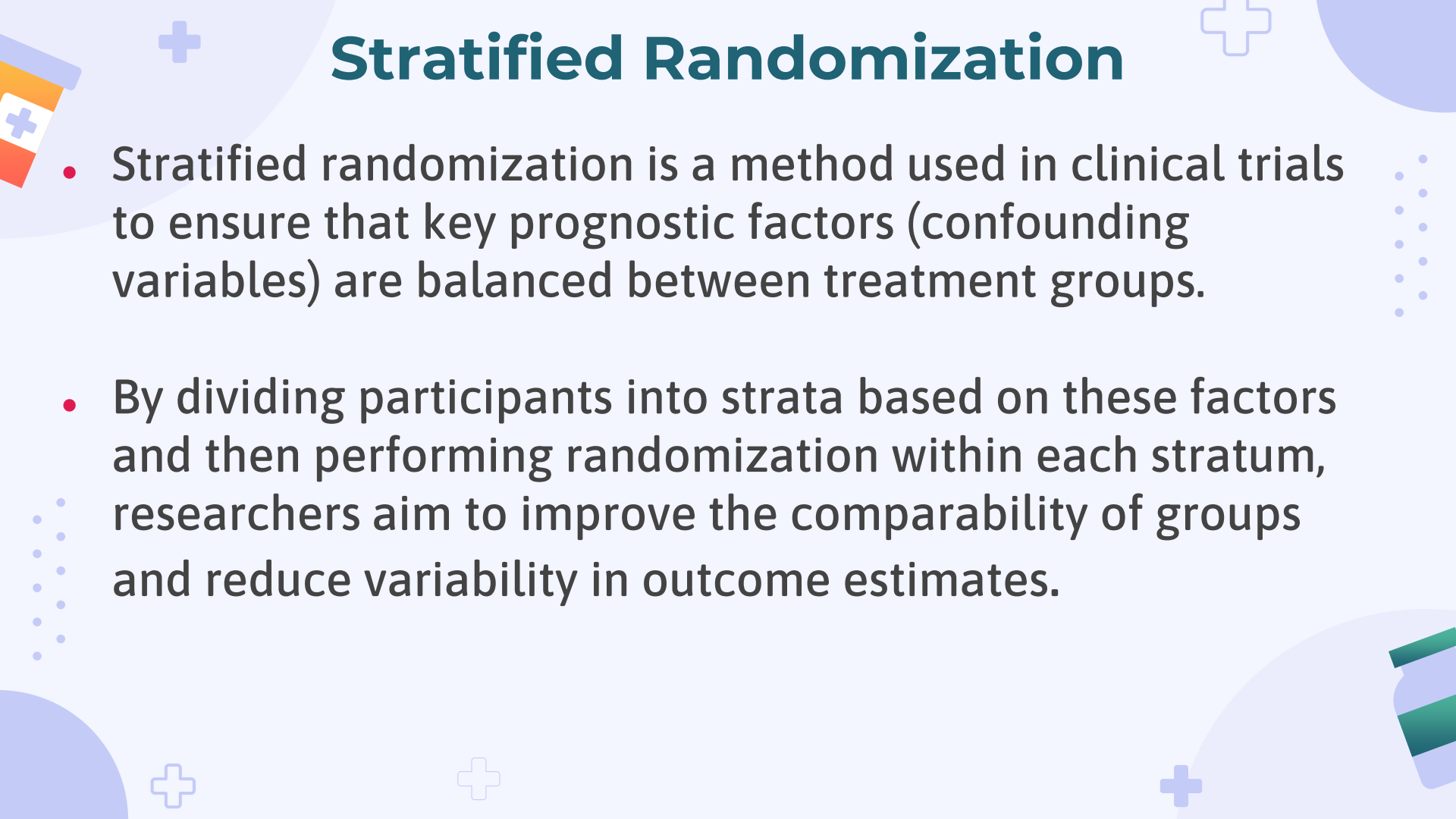


Unequal allocation randomization.

- Unequal allocation randomization is a method used in clinical trials where participants are assigned to study groups (e.g., intervention and control groups) in proportions that deviate from the typical 1:1 ratio.
- Instead, the allocation may follow a ratio such as 2:1, 3:1, or even more extreme ratios.
- This approach can be strategically applied depending on the objectives of the trial.

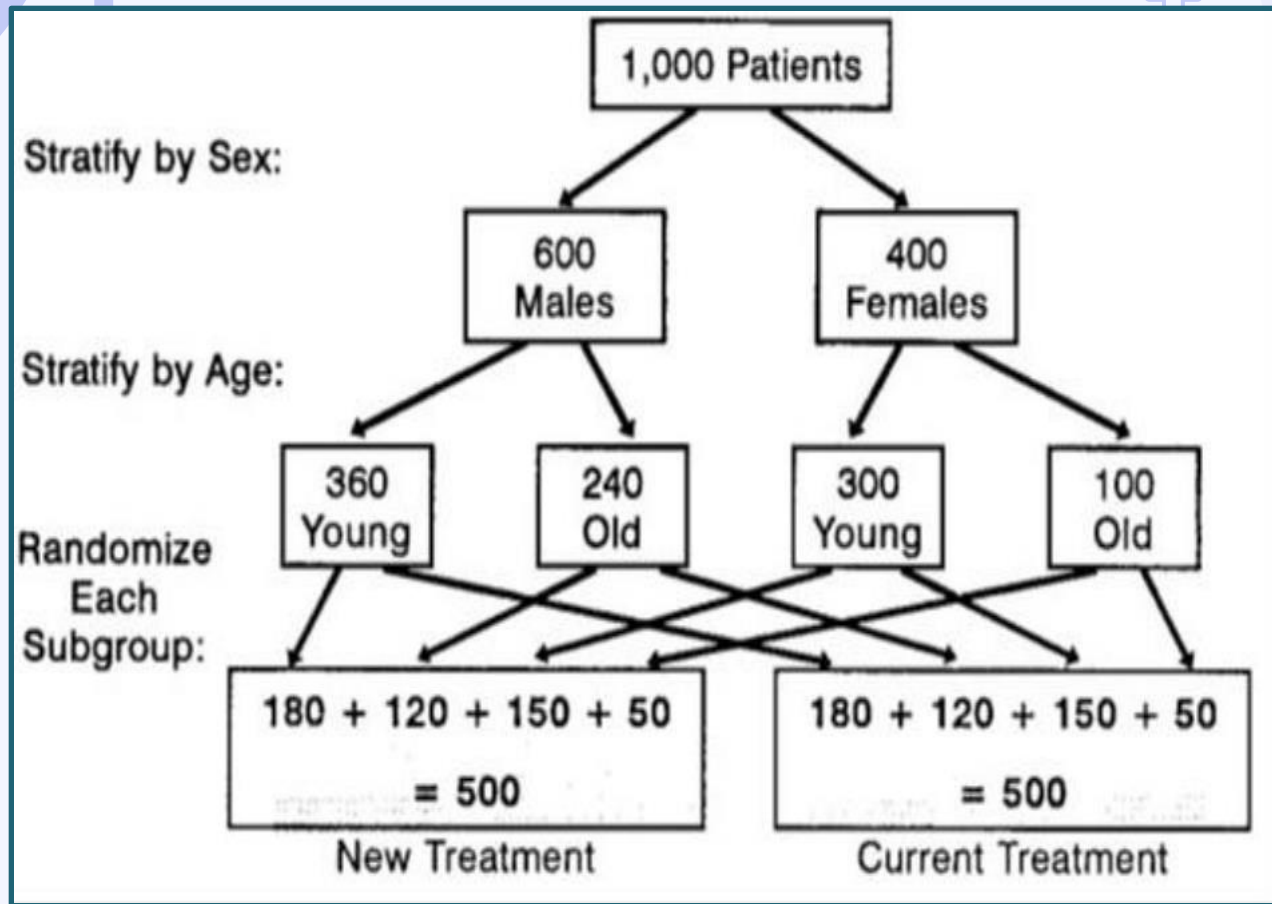
Unequal allocation randomization.

- When two or more treatments under evaluation have a cost difference:
 - Substantial cost savings can be achieved by adopting an unequal randomization ratio such as 2:1, with only a modest loss in statistical power.
 - Another scenario: When one arm saves lives, and the other (such as placebo/medical care only) does not.
 - Generally, a randomization ratio of 3:1 will lose considerable statistical power.



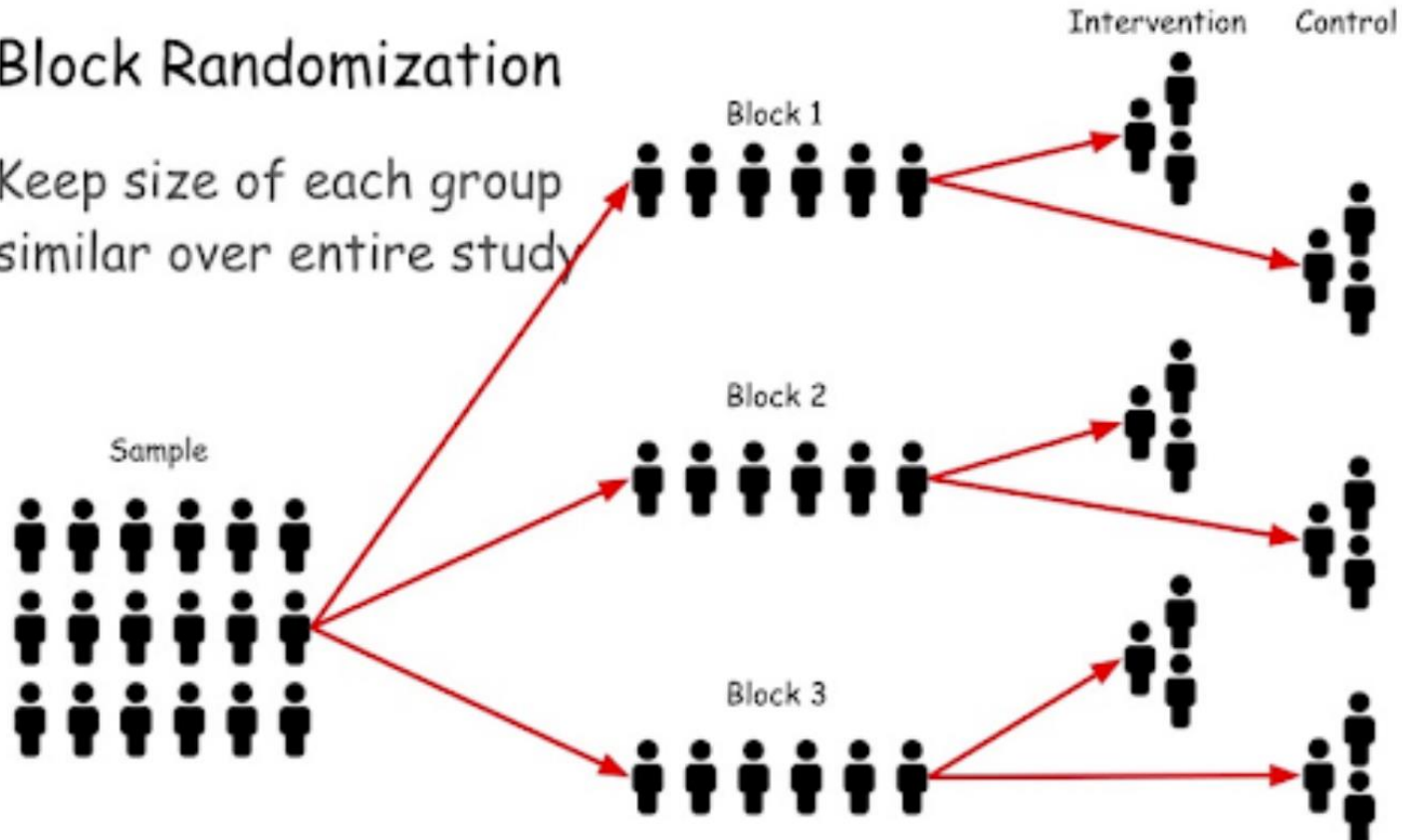
Stratified Randomization

- Stratified randomization is a method used in clinical trials to ensure that key prognostic factors (confounding variables) are balanced between treatment groups.
- By dividing participants into strata based on these factors and then performing randomization within each stratum, researchers aim to improve the comparability of groups and reduce variability in outcome estimates.



Block Randomization

Keep size of each group similar over entire study





Blinding in Clinical Trials



- Blinding keeps participants, clinical trial personnel (e.g., healthcare providers), or assessors (those collecting outcome data) unaware of an assigned intervention so that they are not influenced by that knowledge.



- Blinding prevents bias at several stages of a trial, although its relevance varies according to circumstance.



Types of blinding techniques :

- | | |
|---------------------|---|
| <u>Non-blind</u> | - Investigator and patient know treatment assigned ("open label") |
| <u>Single-blind</u> | - Investigator knows treatment assigned but patient does not (in some cases participants knows and investigator does not) |
| <u>Double-blind</u> | - Neither investigator nor patient knows treatment assigned |
| <u>General view</u> | - Double-blind > single blind > non-blind |



Drug information

By: Dr. Riffat Ibrahim Riffat

What is a Protocol ?

- Gives detailed schedule of tests, procedures, medications, dosages, Sample size and length of the study.
- Describes what types of people may participate in the trial (inclusion and exclusion criteria) .
- Principal Investigator is responsible for assuring that the protocol is strictly followed for each participant

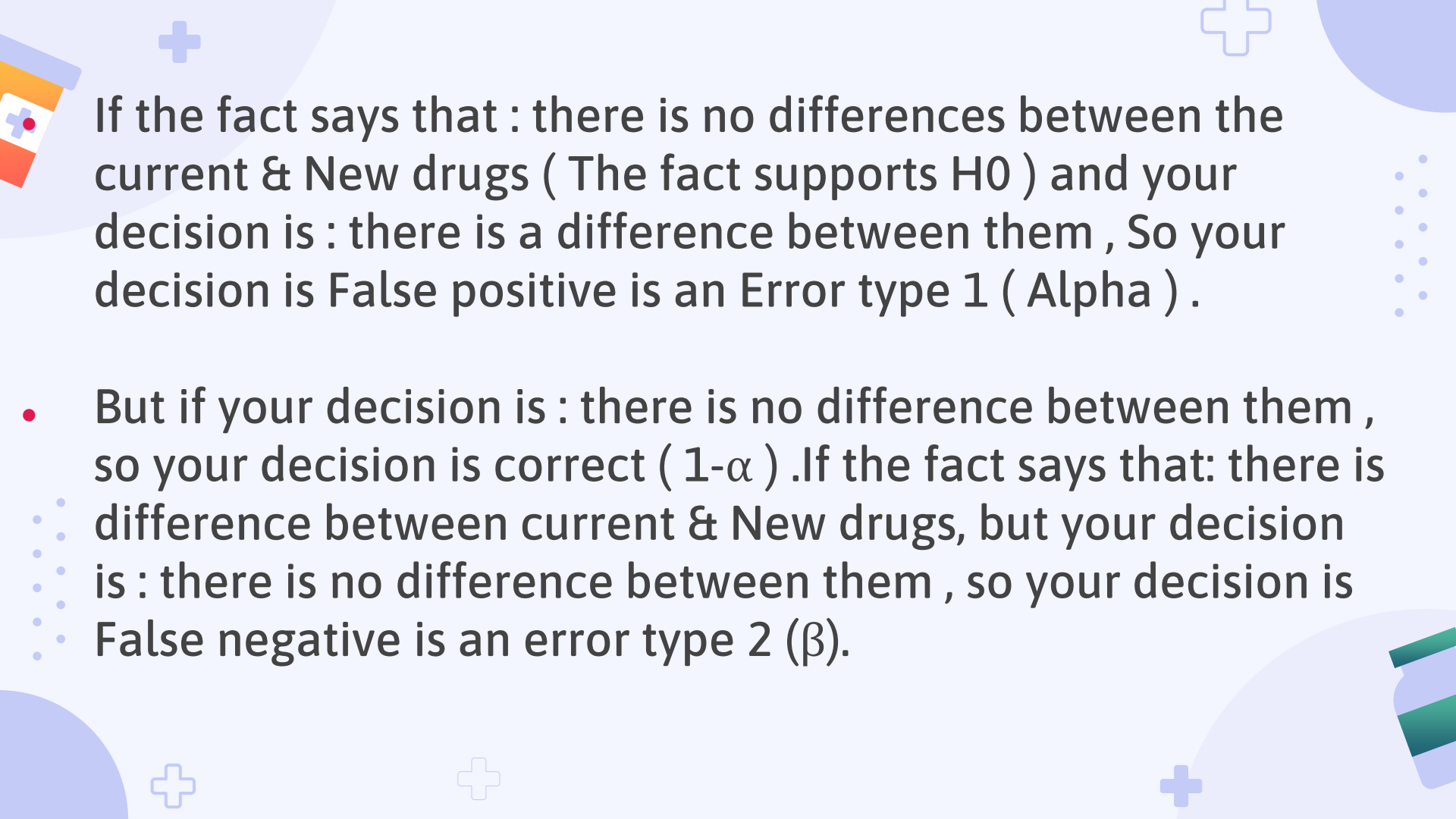
H0 Hypothesis:

- Null Hypothesis Predicts that no differences between two variables.
- So, in a clinical trial we proof the concept relying on the H0 hypothesis.



+ Precision & Power of the study

- Error type 1 (Alpha) : affects precision of the clinical study. The possibility of error type 1 (False positive) to occur is 5% So, The precision is 95% .
- Error type 2 (Beta) : affects power of the clinical study.
- The possibility of error type 2 (False negative) to occur is 20%, So the power is 80% .



If the fact says that : there is no differences between the current & New drugs (The fact supports H_0) and your decision is : there is a difference between them , So your decision is False positive is an Error type 1 (Alpha) .

- But if your decision is : there is no difference between them , so your decision is correct ($1-\alpha$) .If the fact says that: there is difference between current & New drugs, but your decision is : there is no difference between them , so your decision is False negative is an error type 2 (β).

Institutional Review Board (IRB)

- Definition: A committee of specific composition at an affiliated fair, high school, or institution that reviews research plans involving human subjects to determine potential physical and/or psychological risk.

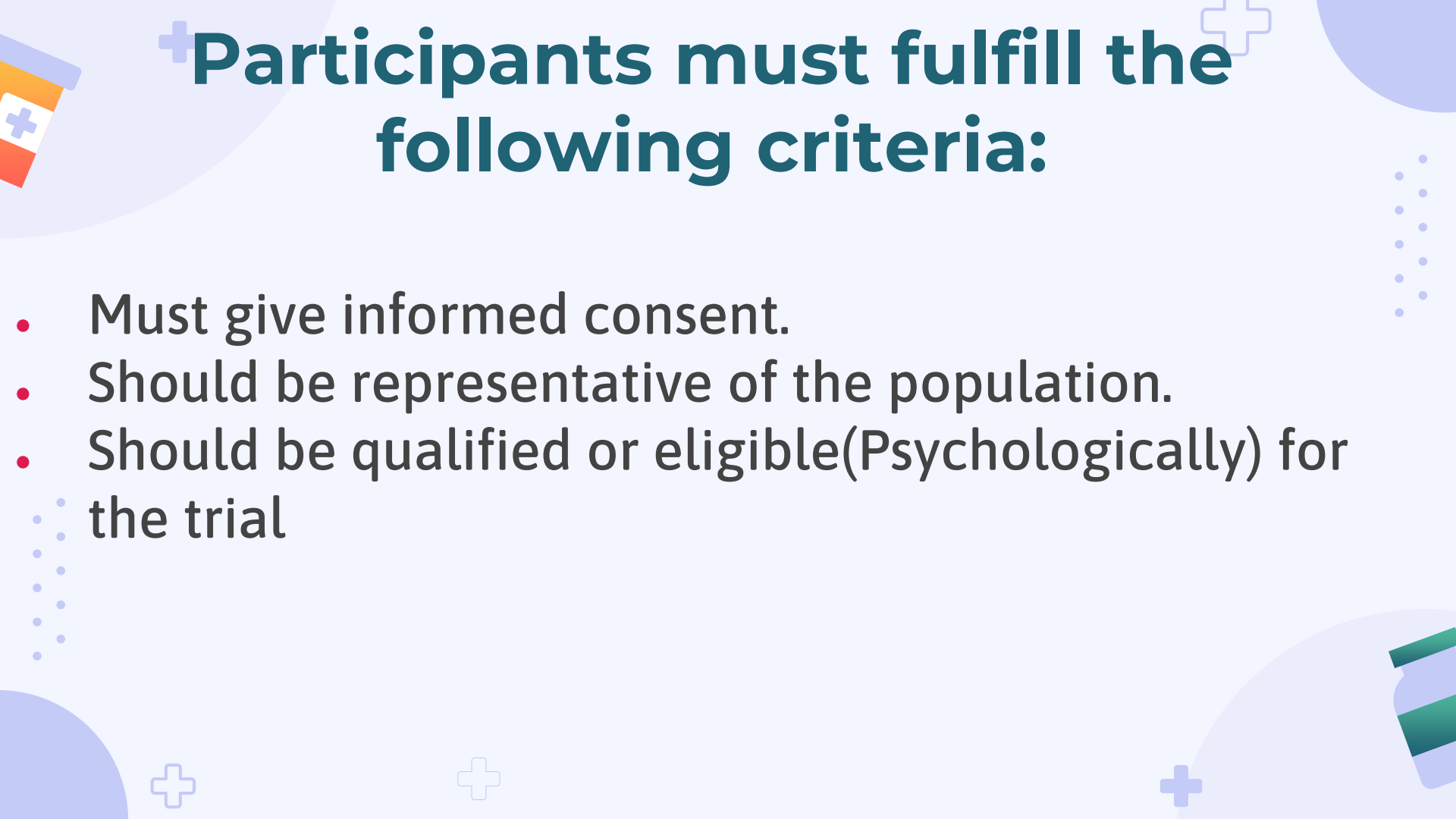


What is informed consent?



- Informed consent means asking the data subject for permission before a process after explaining the activity, for what purpose their data will be used, and how it will be processed.





+ Participants must fulfill the following criteria:

- Must give informed consent.
 - Should be representative of the population.
 - Should be qualified or eligible(Psychologically) for the trial
- 



"Essential Elements of Informed Consent:

- Research description
 - Risks
 - Benefits
 - Procedures
 - Alternatives
 - Confidentiality
 - Contacts
 - Voluntary participation and withdrawal"
- 





Subject dropout and compliance

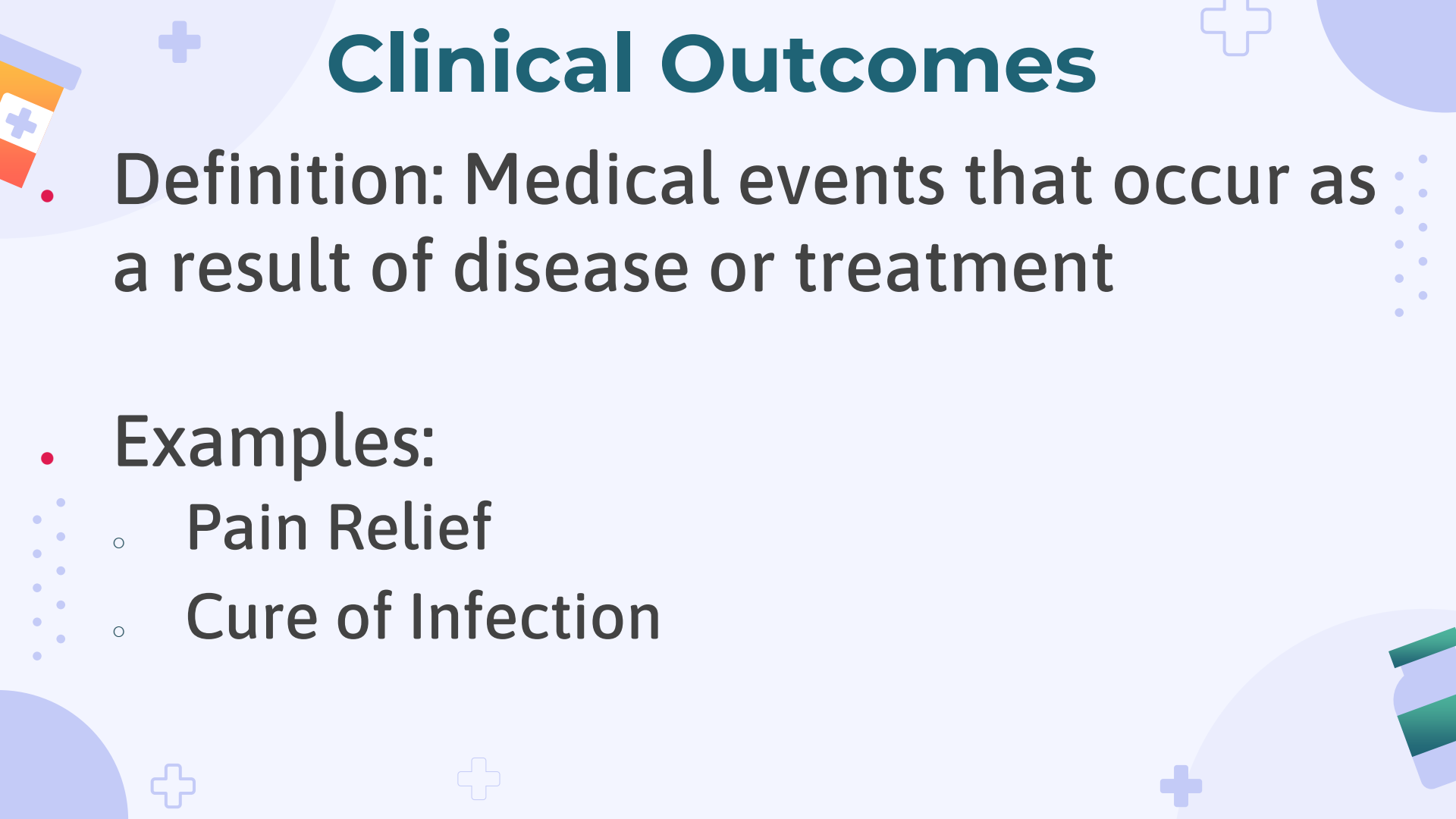


- Not all subjects randomized in a clinical trial will complete the entire duration, at which time they are then termed a study dropout or stated to be lost to follow-up.
 - Reasons for discontinuing study participation :
 - 1-Lack of desire to continue.
 - 2-Subject relocation (i.e., moving to another city).
 - 3-Subject violating study protocol, side effects, and death.
 - 4-Not all subjects will be compliant with the therapy
- 

30%

OF CLINICAL TRIAL
PATIENTS DROP OUT¹





Clinical Outcomes

• Definition: Medical events that occur as a result of disease or treatment

• Examples:

- Pain Relief
- Cure of Infection

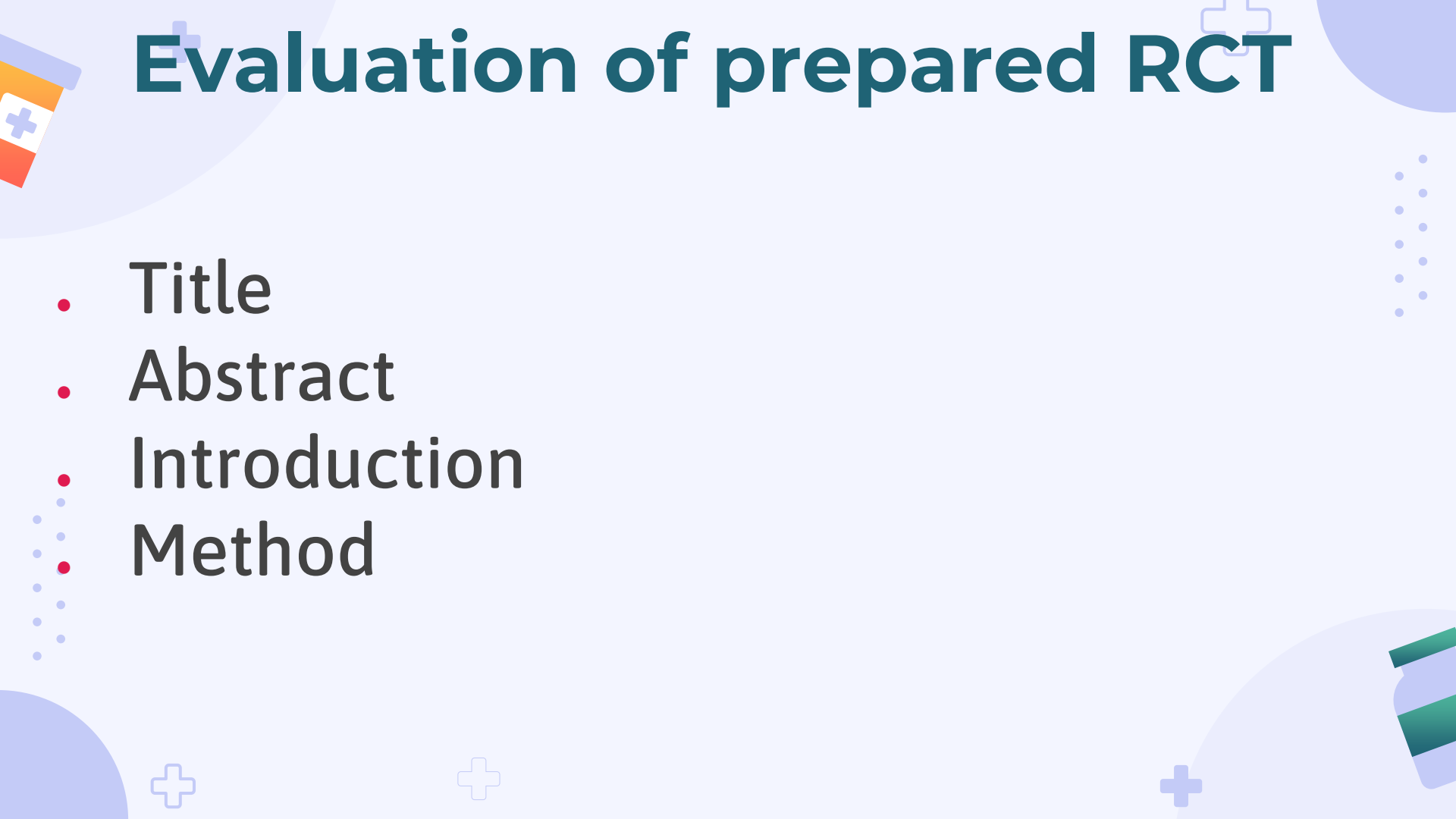
Types of outcomes

- Primary Outcome: is the outcome of interest that has been set by the researcher at the beginning of the study.
- Secondary Outcome: is the additional outcome that occurs during the study.



According to results comparing, can be classified studies in to:

- Positive studies: a favorable result for the therapy understudy, are readily published.
 - Negative studies: a study that does not show favorable results from the new therapy) may not be published. These results may due to the existence of questionable data or poorly designed studies.
- 
- 
- 
- 



Evaluation of prepared RCT

- Title
- Abstract
- Introduction
- Method

Title

- A title should be reflective of the work, unbiased, specific, and concise (i.e., usually ≤ 10 words)
 - not too general or detailed.
 - the title should not be phrased as a question
 - randomized clinical trials should be identified in the title.



Blinding in Clinical Trials



- An example for a biased study title:
 - “Improved bronchodilation with levalbuterol compared with racemic albuterol in patients with asthma.”
- suggested unbiased title for this trial is:
 - “Comparison of levalbuterol and racemic albuterol bronchodilation in patients with asthma: a randomized clinical trial.”



Abstract:

- Abstracts include information:
 - the article objective, methods, results, and conclusions. usually contain ≤ 500 words.

Types of abstracts:

- a. structured
- b. non-structured

Abstract:

- Advantages of Structured abstracts :
 - more informative
 - easier to read
 - generally preferred by readers.

Introduction

- should be discussing the study rationale
- clearly stated research purpose
- should explain how the clinical trial will overcome the shortcomings of the prior research

Methods

Methods should include:

- The type of subjects enrolled
- The comparative therapy description
- Outcome measures
- Statistics.

Good research

- Discusses key decisions researchers faced when structuring their analysis.
- It may perform multiple analyses using alternative approaches, and compares their results.
- It demands multiple types of evidence to reach a conclusion. It does not assume that association (things occur together) proves causation (one thing causes another).

Bad research

- Often uses accurate data but manipulates the information to support a particular conclusion.
- Questions can be defined, statistics selected and analysis structured to reach a desired outcome.
- Alternative perspectives and data can be ignored or distorted

The investigators

- Need to be properly trained and have active practice experience in the area of study.
- The site where the clinical trial was conducted (have the resources to properly and completely perform the necessary study methods).
- Persons involved with the study need to be ethical and responsible to protect patients enrolled in the study



Drug information

By: Dr. Riffat Ibrahim Riffat



+ Adverse drug reaction (W.H.O Definition)

- Adverse drug reaction is defined as any response to a drug that is noxious and unintended and that occurs at doses used in man for prophylaxis, diagnosis, or therapy of disease or for modification of physiological function.
- 

Side-effects

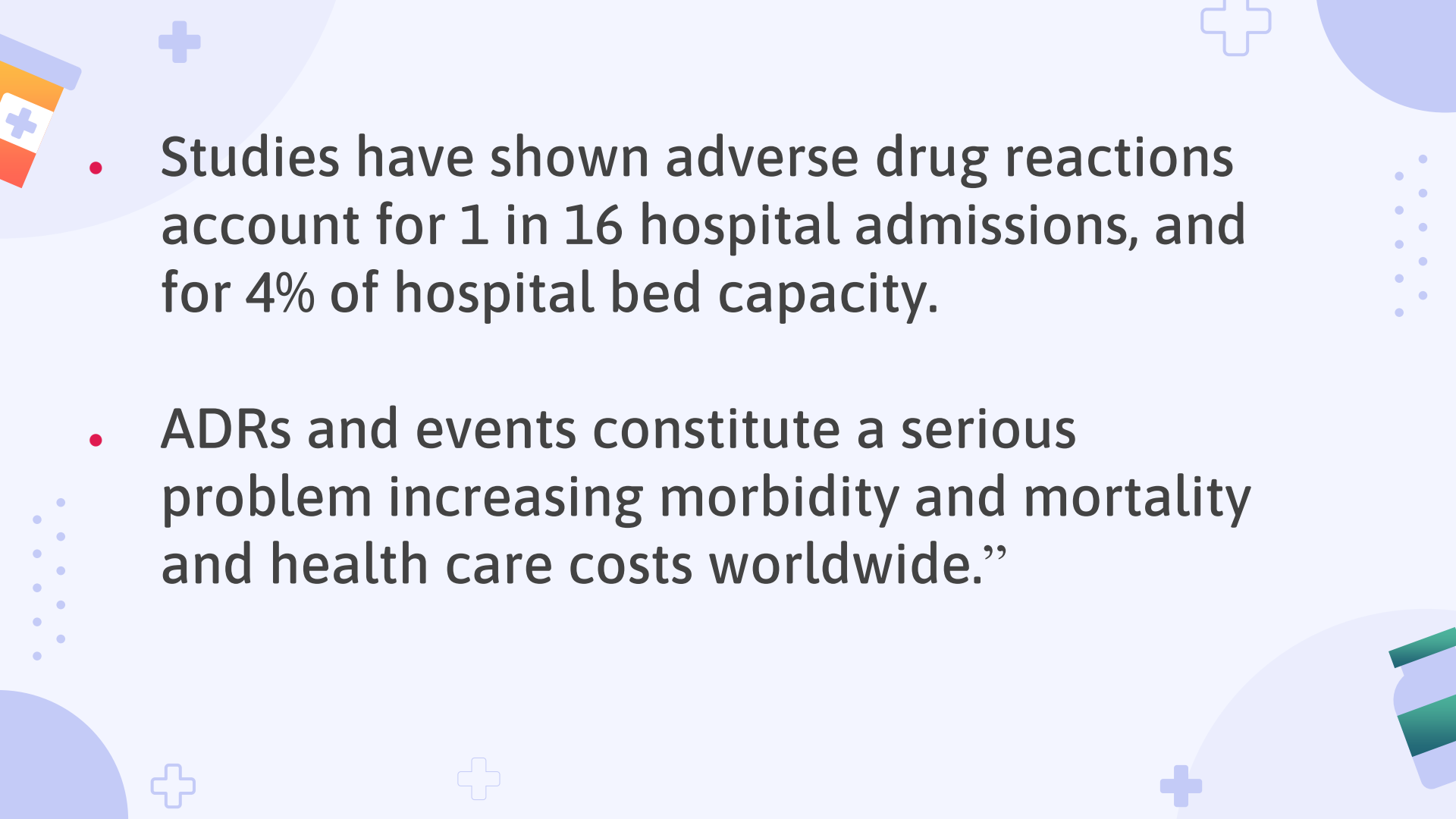
- A consequence of the pharmacological mechanism of action of the drug - usually due to the presence of receptors in a number of tissues or lack of receptor specificity.
- Examples include drowsiness with the older antihistamines, constipation with opioids, etc.
- In some situations, a 'side-effect' can be a therapeutic effect, e.g., antihistamines for sedation, opioids for diarrhea

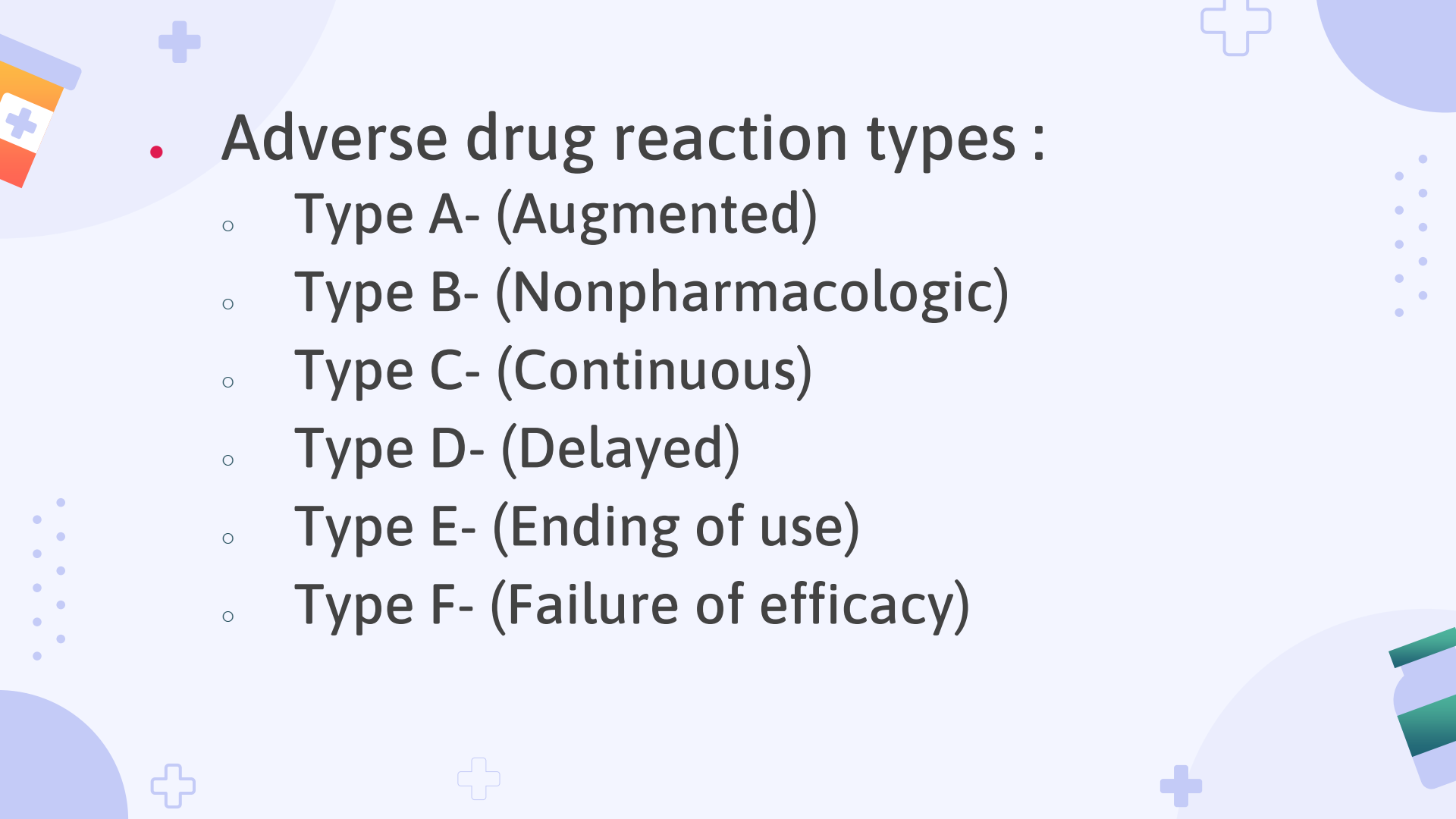
The burden of ADRs

• Causality:

- The probability that a particular medicine is responsible for an isolated effect or ADR.

• Adverse drug reactions are frequently serious enough to result in admission to hospital. It is well recognized that adverse drug reactions place a significant burden on the health service

- 
- Studies have shown adverse drug reactions account for 1 in 16 hospital admissions, and for 4% of hospital bed capacity.
 - ADRs and events constitute a serious problem increasing morbidity and mortality and health care costs worldwide.”

- 
- Adverse drug reaction types :
 - Type A- (Augmented)
 - Type B- (Nonpharmacologic)
 - Type C- (Continuous)
 - Type D- (Delayed)
 - Type E- (Ending of use)
 - Type F- (Failure of efficacy)

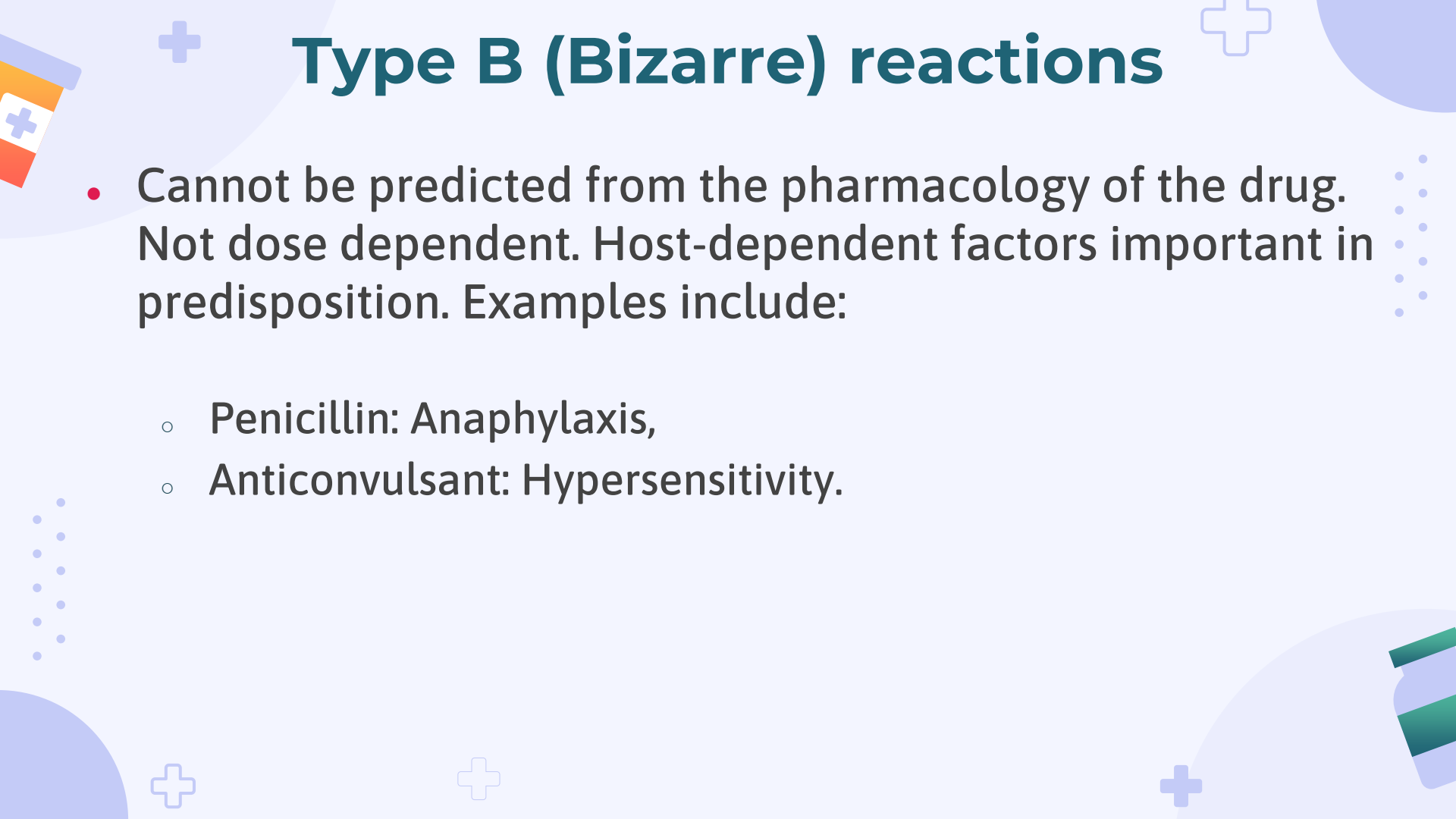


+ Type A (Augmented) reactions



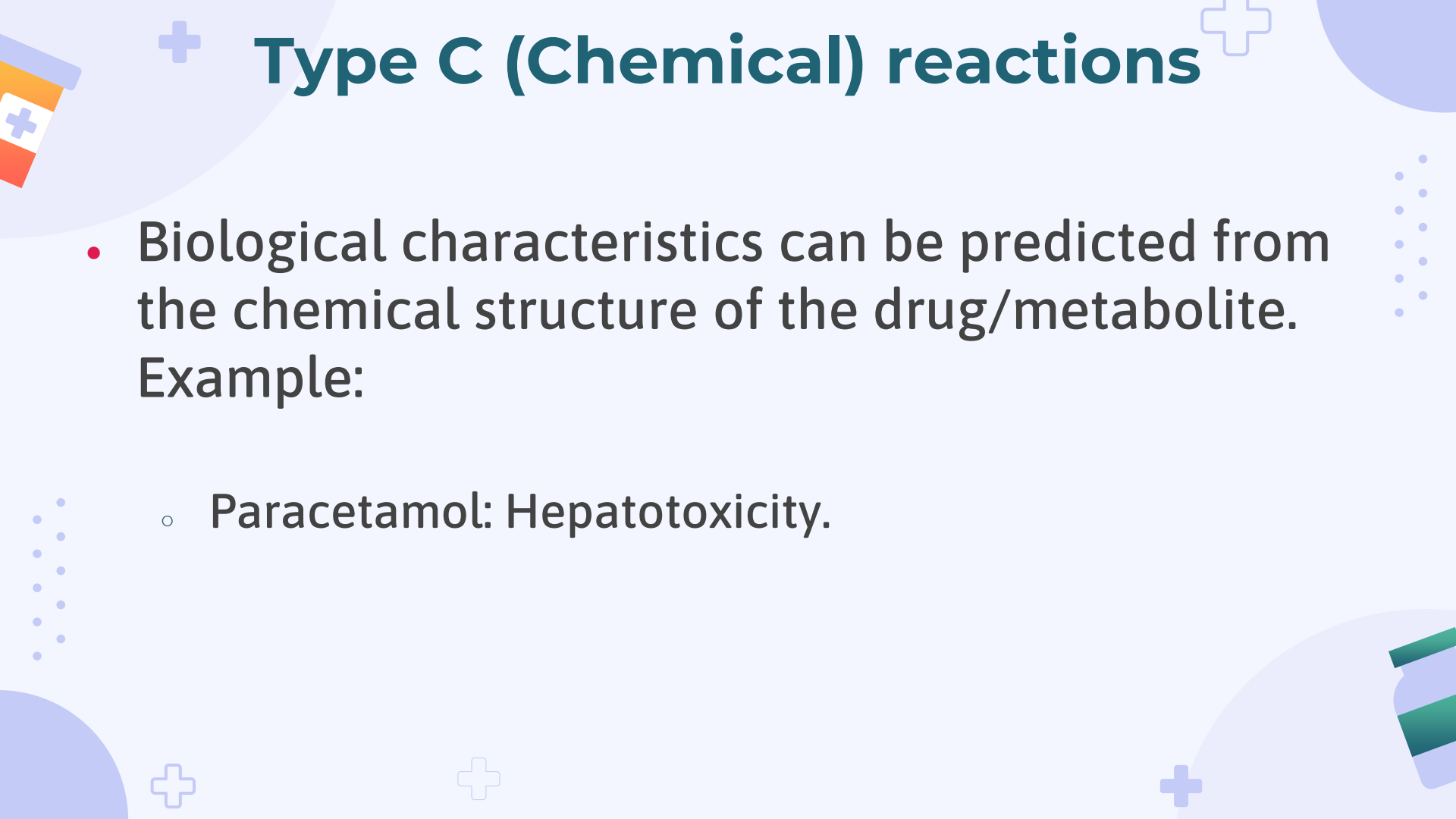
- Reactions which can be predicted from the known pharmacology of the drug. Dose dependent, can be alleviated by a dose reduction. Examples include:
 - Anticoagulants: Bleeding,
 - Beta blockers: Bradycardia,
 - Nitrates: Headache,
 - Prazosin: Postural hypotension."
- 





Type B (Bizarre) reactions

- Cannot be predicted from the pharmacology of the drug. Not dose dependent. Host-dependent factors important in predisposition. Examples include:
 - Penicillin: Anaphylaxis,
 - Anticonvulsant: Hypersensitivity.



+ Type C (Chemical) reactions

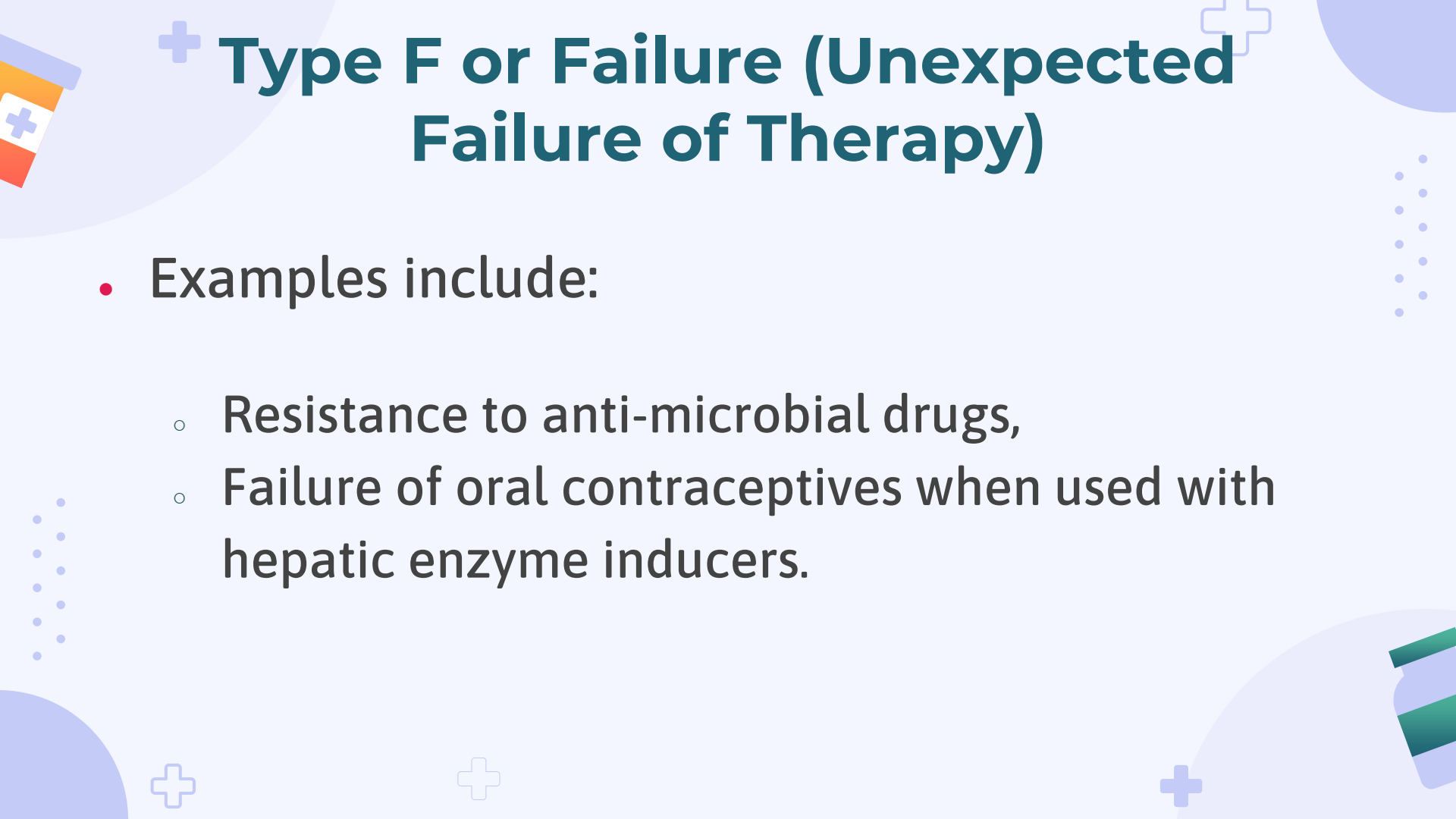
- Biological characteristics can be predicted from the chemical structure of the drug/metabolite.
Example:
 - Paracetamol: Hepatotoxicity.



Type E (End of treatment) reactions

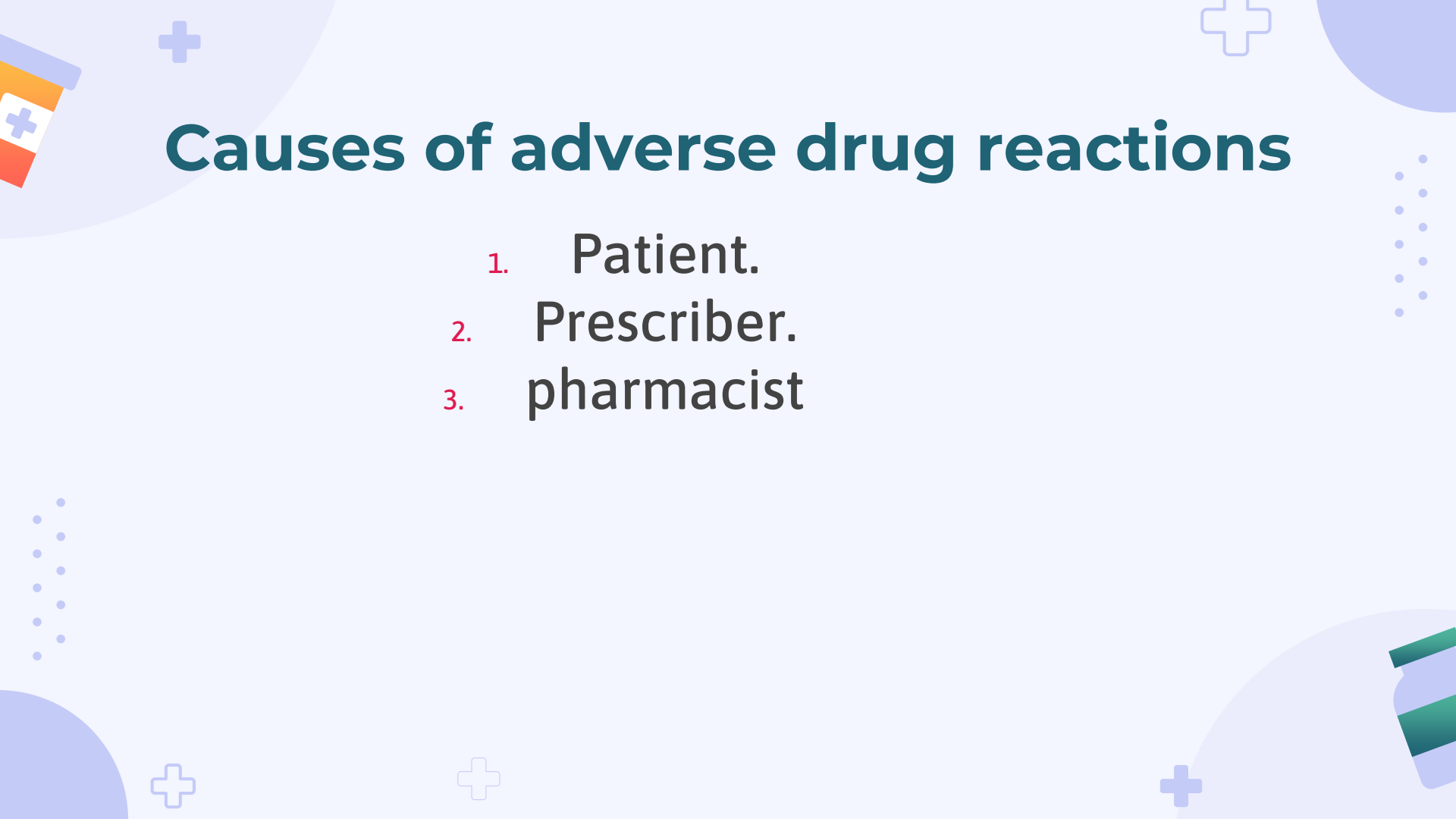
- Occur on withdrawal, especially when the drug is stopped abruptly. Examples include:
 - Phenytoin withdrawal: Seizures,
 - Steroid withdrawal: Adrenocortical insufficiency.
- 





+ Type F or Failure (Unexpected Failure of Therapy)

- Examples include:
 - Resistance to anti-microbial drugs,
 - Failure of oral contraceptives when used with hepatic enzyme inducers.



Causes of adverse drug reactions

1. Patient.
2. Prescriber.
3. pharmacist



Patient Risk Factors for Adverse Drug Reactions



- Female gender
 - Serious illness
 - Renal insufficiency
 - Liver disease
 - Poly pharmacy
 - HIV infection
 - Alcoholism
 - Genetics
- 
- 
- 
- 



+ Prescriber Risk Factors for Adverse Drug Reactions



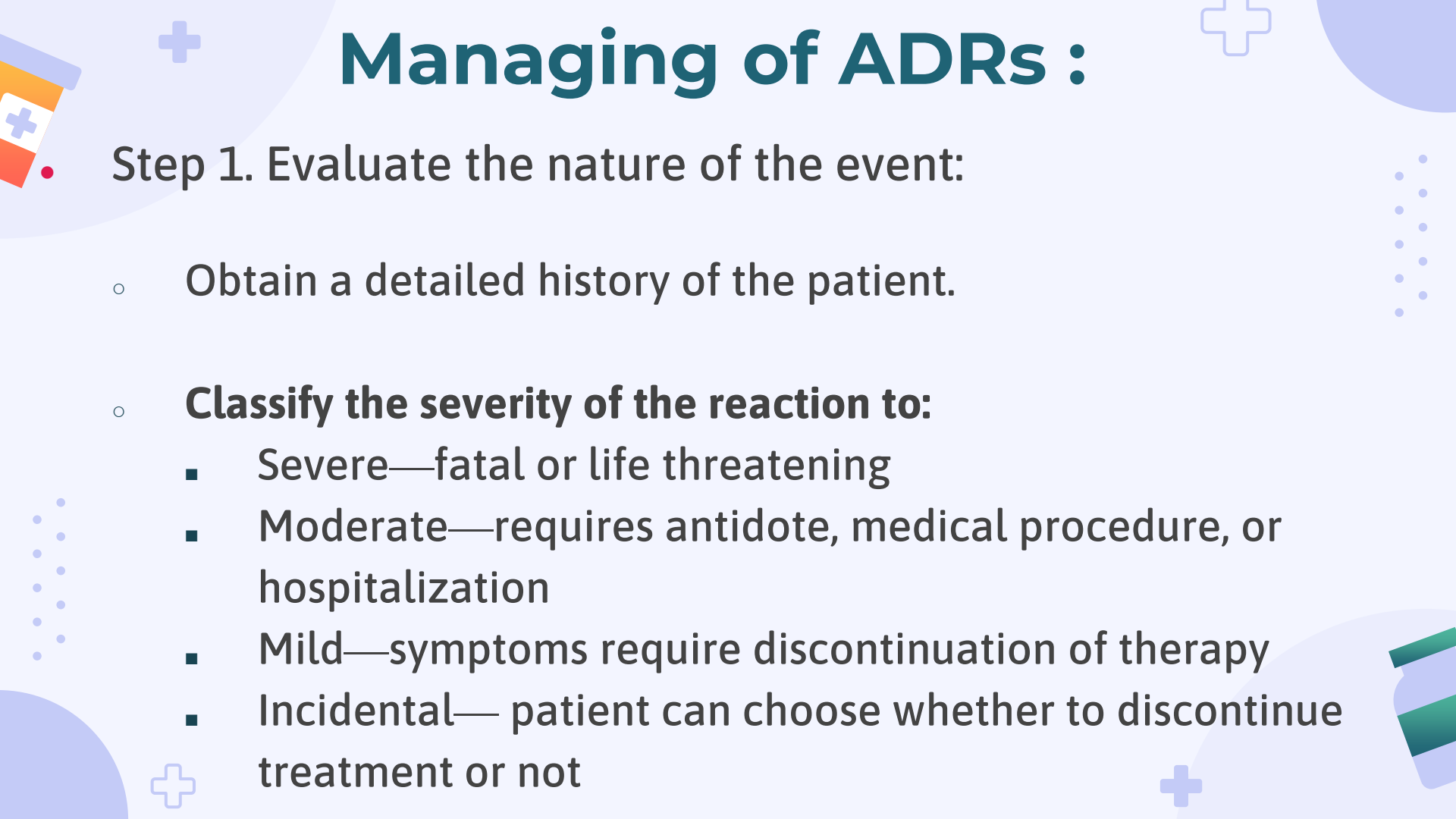
- ADR may occur if drug used for inappropriately long time (Type C), at a critical phase in gestation (Type D) or is abruptly (Type E) or given with other drugs (Drug-drug interactions).
- 
- 

Pharmacovigilance

- Refers to the process of continuous monitoring for unwanted effects and other safety related aspects of marketed drugs.

Clinical Outcomes

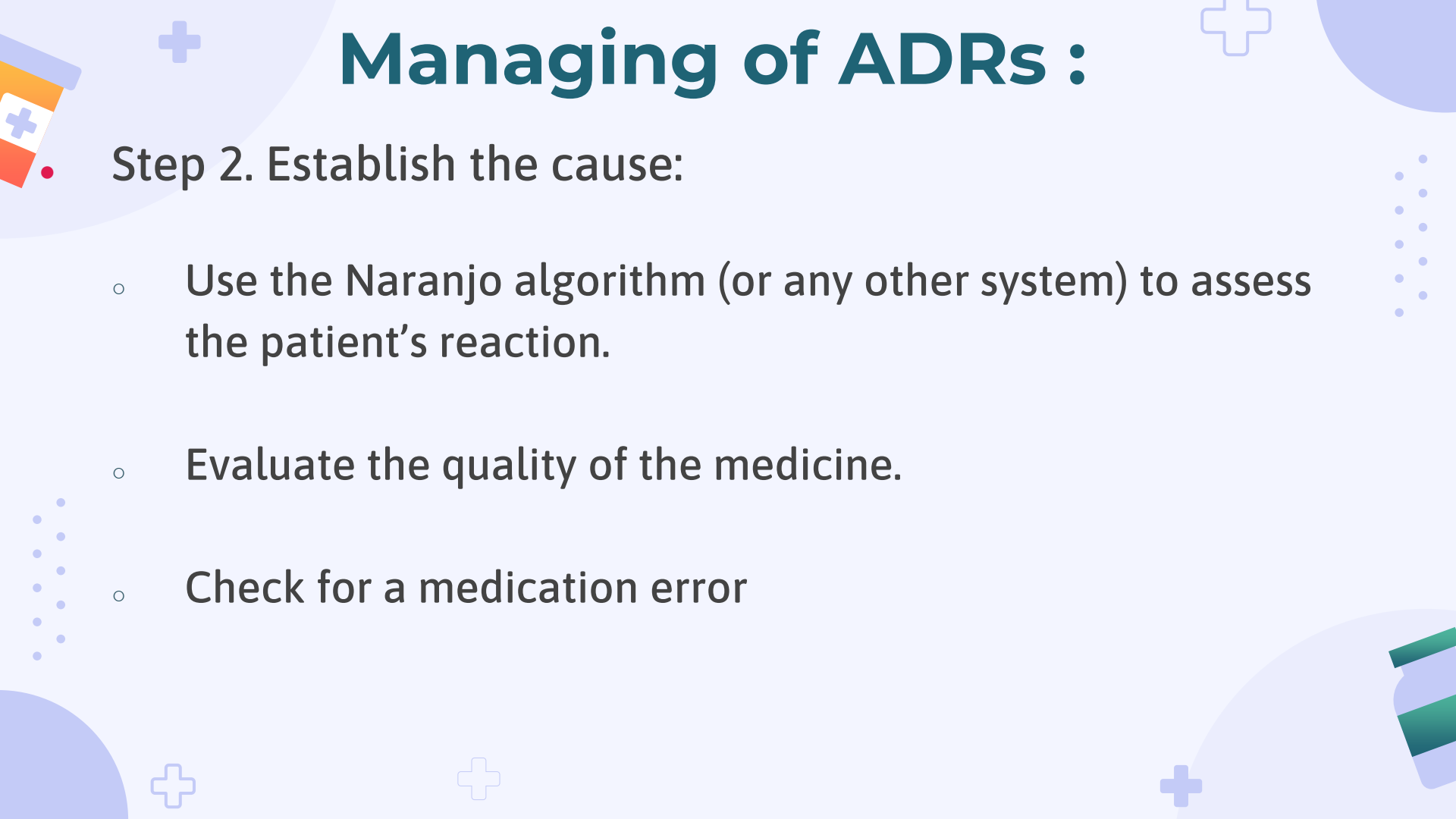
- Identifying and Managing ADRs medicine Safety:
 - Premarketing clinical trials
 - Animal studies, human studies—Phases I, II, III
 - Cannot identify ADRs with incidence $< 1\%$
- Post marketing surveillance
 - • Spontaneous reporting
 - • Post marketing clinical trials—Phase IV
 - • Other methods—observational studies, meta-analysis, case reports
 - • Determining causality
 - • Actions taken to manage new ADR



Managing of ADRs :

• Step 1. Evaluate the nature of the event:

- Obtain a detailed history of the patient.
- **Classify the severity of the reaction to:**
 - Severe—fatal or life threatening
 - Moderate—requires antidote, medical procedure, or hospitalization
 - Mild—symptoms require discontinuation of therapy
 - Incidental— patient can choose whether to discontinue treatment or not



Managing of ADRs :

• Step 2. Establish the cause:

- Use the Naranjo algorithm (or any other system) to assess the patient's reaction.
- Evaluate the quality of the medicine.
- Check for a medication error



Managing of ADRs :

• Step 3. Take corrective and follow-up action.

○ **For Severe ADRs:**

- Educate and monitor prescribers.
- Change the formulary or standard treatment guideline.
- Notify drug regulatory authorities and manufacturers.

○ **All ADRs:**

- Educate and warn patients



Implications for drug therapeutic committee (DTC)

Surveillance of ADRs :

- Monitoring and managing ADRs requires setting up surveillance systems:
 - Use of local surveillance (tracking and reporting) system run by the DTC
 - Use of standardized reporting forms
 - Analysis of reported ADRs to be done by selected DTC committee member
 - Reporting of serious and recurring ADRs to regulatory authorities and manufacturers
- 



DTC's (Drug therapeutic committee) Role in Preventing ADRs:

- Analyze reported ADRs.
 - Discuss changes in the formulary or standard treatment guidelines for significant or recurring problems with ADRs.
 - Identify medicines on the formulary that are “high risk” and should be monitored closely by physicians and pharmacists
- 

Drug information

By: Dr. Riffat Ibrahim Riffat

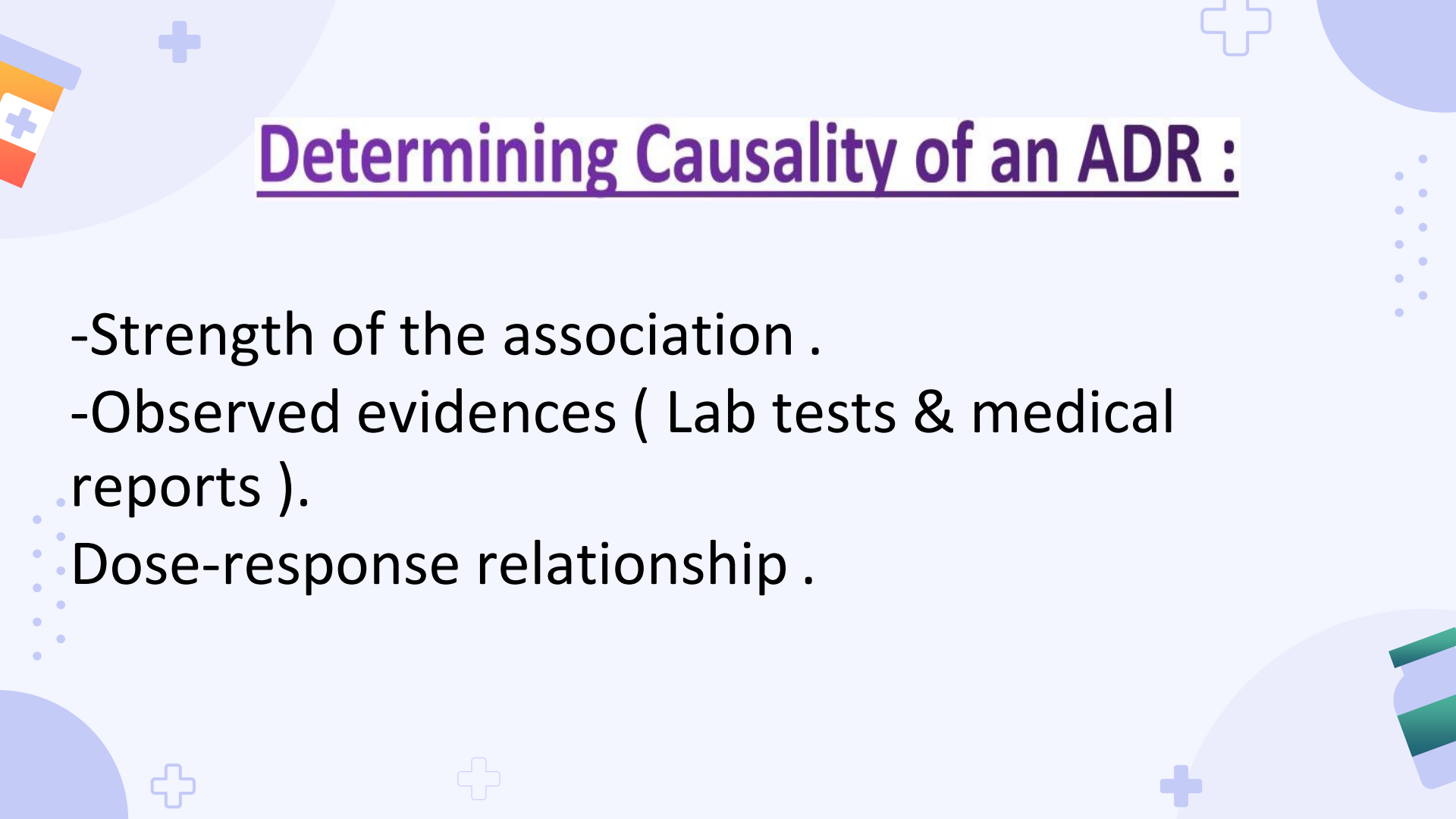


- **Classifying Causality of an ADR**

- *Definite causality—when a clinical event (including laboratory test) occurs in a plausible time relationship to medicine administration and cannot be explained by concurrent disease or other medicines or chemicals; re- administration of the medicine causes a similar reaction
- *Probable (or likely) causality—when a clinical event occurs with a reasonable time sequence to medicine administration and is unlikely to be due to any concurrent disease or other medicine administration

- **Classifying Causality of an ADR**

- **Possible causality**—when a clinical event occurs with a temporally time sequence to medicine administration and could be explained by concurrent disease or other medicine administration.
- **Doubtful or Unlikely causality:**—when a clinical event occurs in temporal relationship to medicine administration that makes a causal relationship improbable, and when other medicines, chemicals, or underlying disease provide plausible explanations.



Determining Causality of an ADR :

- Strength of the association .
- Observed evidences (Lab tests & medical reports).
- Dose-response relationship .

Methods for assessing ADRs causality

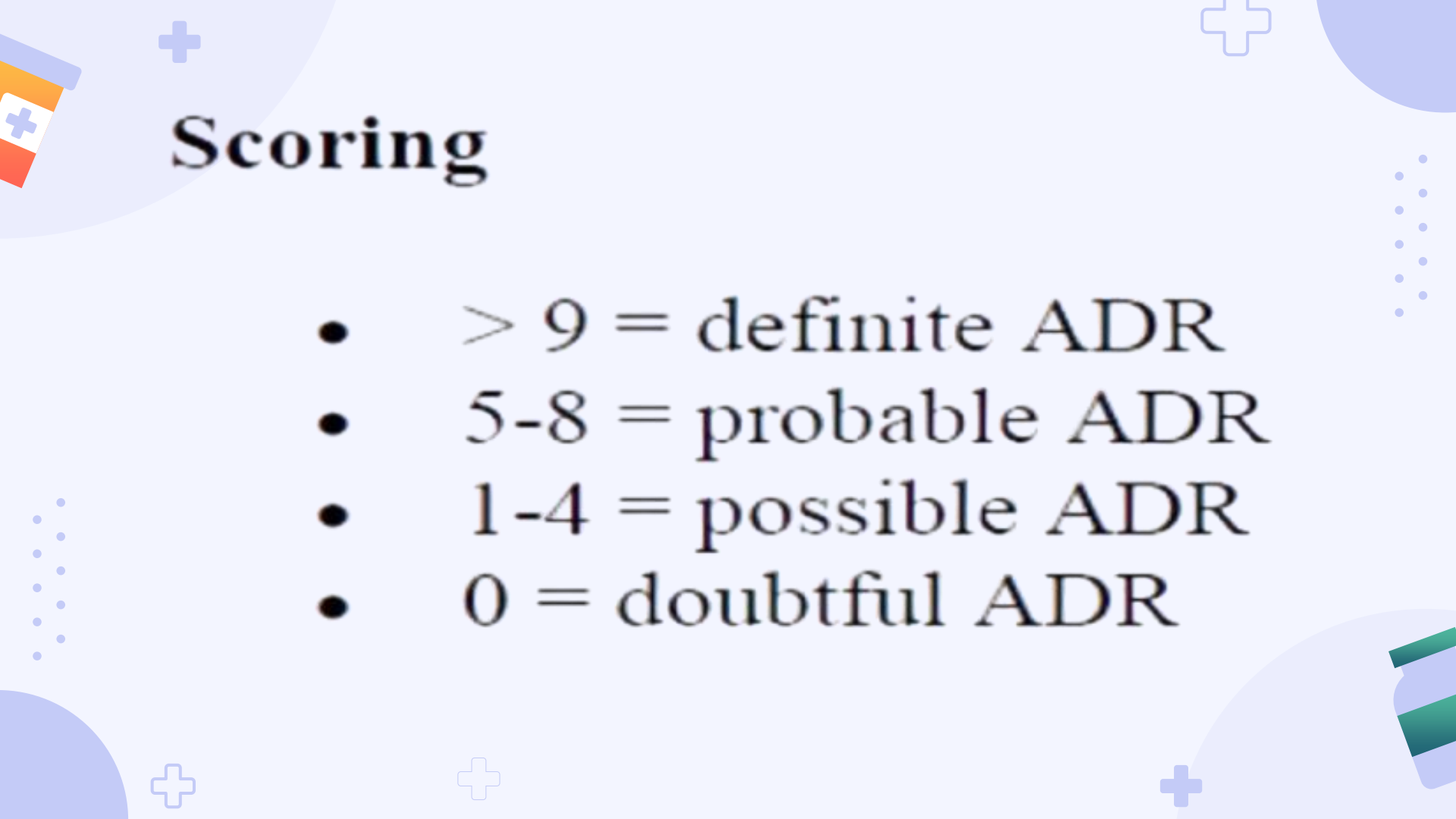
- **1. Expert judgment/Global introspection:**
 - This method involves the individual assessment of the event by a health care practitioner, based on his or her clinical knowledge and experience without using any form of standardized tool.
- **2. Algorithm:**
 - This method uses specific questions to assign a weighted score that helps determine the probability of causality in a given reaction .

3. advanced statistical methods.

Naranjo Adverse Drug Reaction Probability Scale

Question	Yes	No	Do Not Know	Score
1. Are there previous <i>conclusive</i> reports on this reaction?	+1	0	0	
2. Did the adverse event appear after the suspected drug was administered?	+2	-1	0	
3. Did the adverse reaction improve when the drug was discontinued or a <i>specific</i> antagonist was administered?	+1	0	0	
4. Did the adverse event reappear when the drug was re-administered?	+2	-1	0	
5. Are there alternative causes (other than the drug) that could on their own have caused the reaction?	-1	+2	0	

6. Did the reaction reappear when a placebo was given?	-1	+1	0	
7. Was the drug detected in blood (or other fluids) in concentrations known to be toxic?	+1	0	0	
8. Was the reaction more severe when the dose was increased or less severe when the dose was decreased?	+1	0	0	
9. Did the patient have a similar reaction to the same or similar drugs in <i>any</i> previous exposure?	+1	0	0	
10. Was the adverse event confirmed by any objective evidence?	+1	0	0	



Scoring

- > 9 = definite ADR
- 5-8 = probable ADR
- 1-4 = possible ADR
- 0 = doubtful ADR

MEDICATION ERRORS



Definition:

Any preventable event that may cause or lead to inappropriate medication use or patient harm while the medication is in control of the health care professional, or patient.



*RISK FACTORS FOR ERRORS

- Lack of knowledge (Physician , pharmacist and nurse)
- Unclear labeling of drug.
- Misidentification of patient.
- Mental lapses .
- Verification errors .

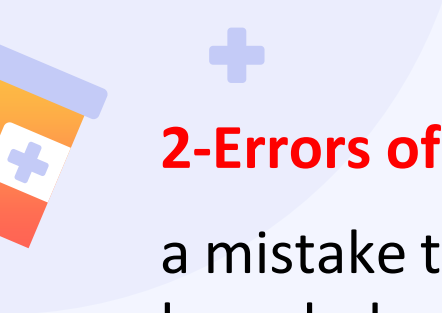


Causes of errors:

1-Errors of Omission:

- a mistake that consists of not doing something you should have done .
- Errors related to prescription, drug dispensing by pharmacist,
drug administration by medical personnel like nurses & patient themselves.





2-Errors of Commission:

a mistake that consists of doing something wrong due to Lack of knowledge .

Types of errors:

1-Prescribing errors.

2-Dispensing errors.

3-Administration errors.



Prescribing errors:



Incorrect drug
selection for a
patient

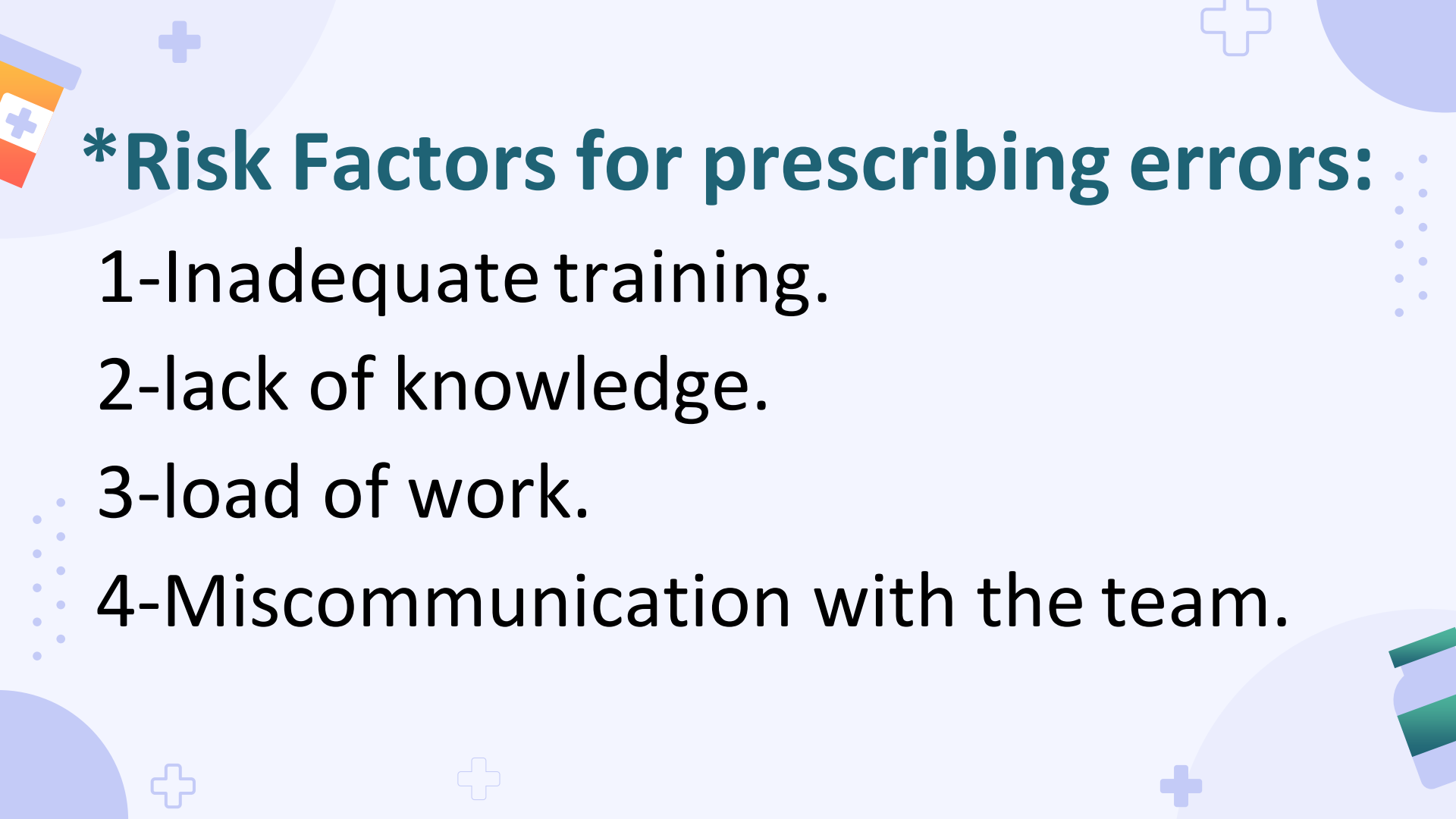


Errors in
quantity,
indication



Prescribing of a
contraindicated
drug





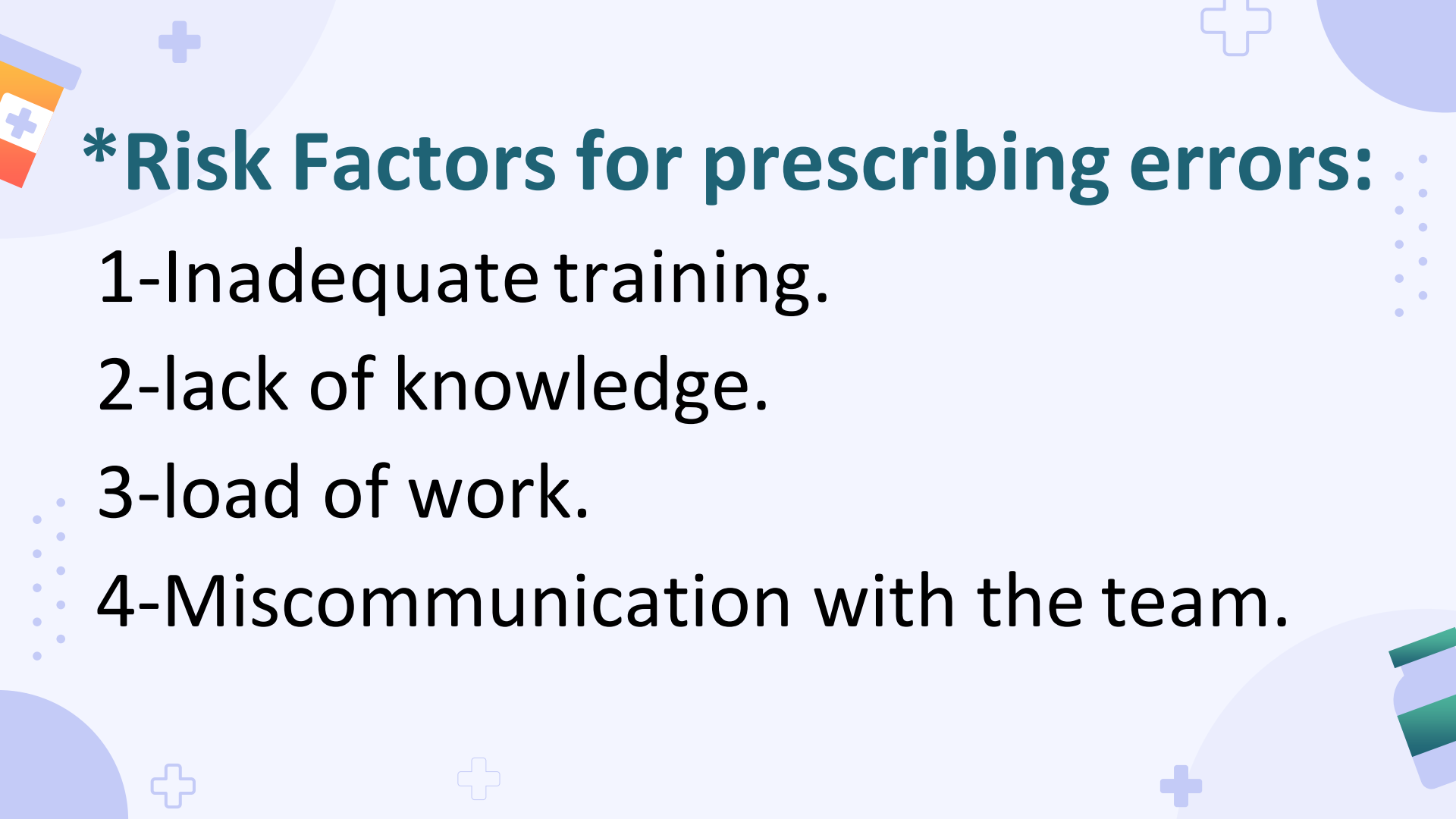
***Risk Factors for prescribing errors:**

1-Inadequate training.

2-lack of knowledge.

3-load of work.

4-Miscommunication with the team.



Bad Handwriting Responsible for Up to 61% of Prescription Errors

Jane
1234 Huang Di Lane
Anywhere, USA 12345 • 208-123-4567

Date
Secondary Dr.

Patient's Name
Primary Dx.

Instructions:

R. Cephalexin 250mg XR
Tid. qid. ad.

R. Cephalexin 250mg XR
Tid. qid. ad.

Take ___ x's per day/month
Frequency ☐ morning ☐ noon ☐ night
Do ___ x's per day/month
Meals ☐ before ☐ during ☐ after
☐ Tuesday ☐ Wednesday ☐ Thursday ☐ Friday ☐ Saturday



Commonplace

www.commonplacefacts.wordpress.com

Reducing dispensing errors :

- 1- Electronic prescribing .
- 2-Unit dose.
- 3- Computerizing.
- 4-Separating drugs with similar name or appearance.

Prescribe Saxenda® and NovoFine® 32G Tip needles through your EHR system—including the 4-week dose escalation



ePrescribing Information

Saxenda®

- Form/strength: **18 mg/3 mL**
- Quantity: **15 mL**
- Dosage form: **Solution, NDC 0169-2800-15**

NovoFine® 32G Tip needle

- Quantity: **1 box (#100)**
- Dosage form: **Disposable needles, NDC 0169-1851-89**

STARTING ON SAXENDA®: disp: 5 pens

sig:	Week 1	0.6 mg SC once daily x 7 days
	Week 2	1.2 mg SC once daily x 7 days
	Week 3	1.8 mg SC once daily x 7 days
	Week 4	2.4 mg SC once daily x 7 days
	Week 5	3 mg SC once daily x 7 days

STAYING ON SAXENDA®: disp: 5 pens

sig: 3 mg SC once daily



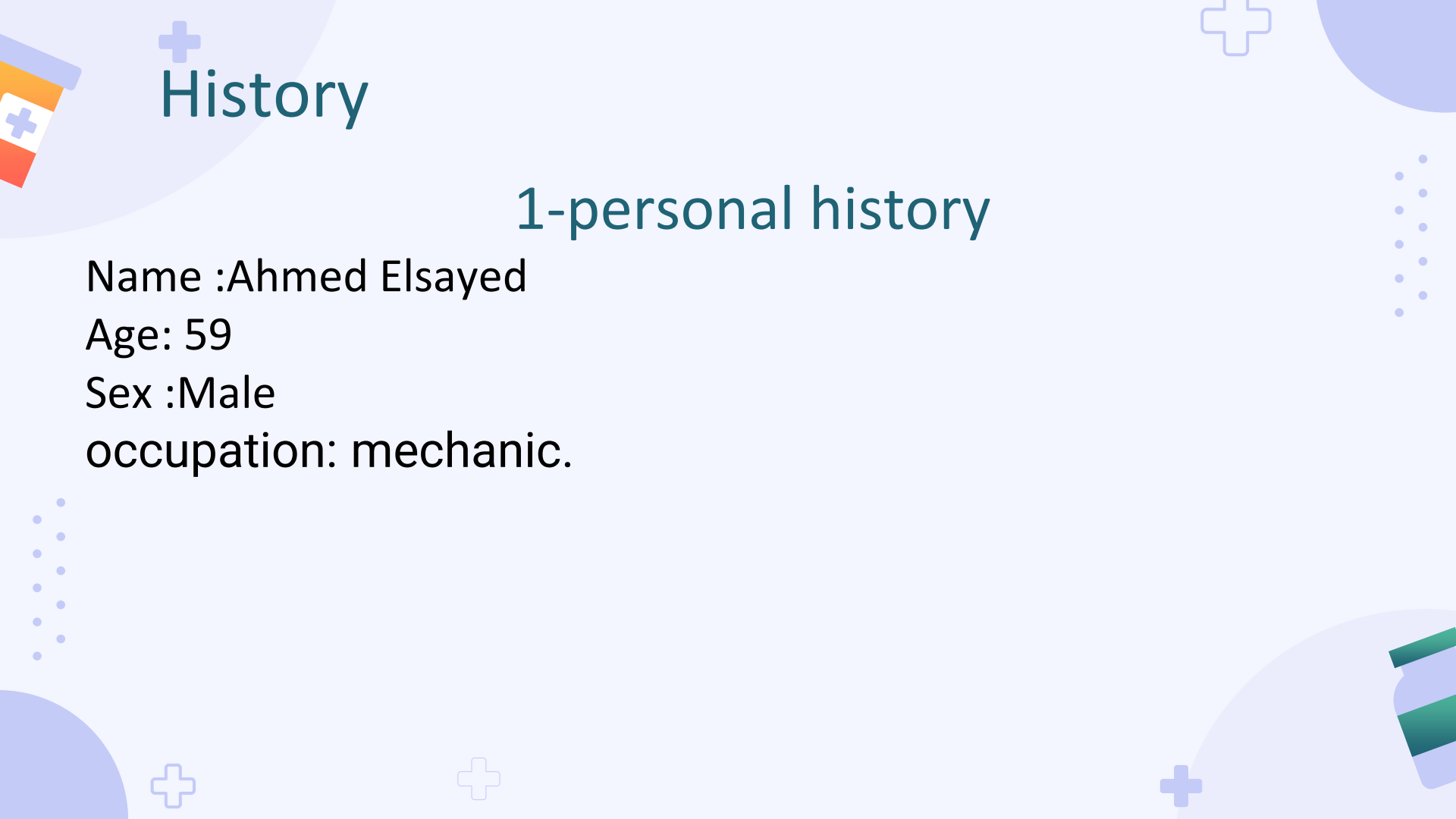
Case study Trauma

The background is a light purple color with a large, faint, light purple stethoscope graphic centered behind the text. Various medical icons are scattered around: a red and white pill bottle in the top left, a clipboard with a checklist in the top right, a red heart with a white cross in the bottom left, and a green and white pill bottle in the bottom right. There are also several small white plus signs and a vertical dotted line on the right side.

Case

2-Case Study

Ahmed Elsayed have 59 years old, job mechanic, with no history of shoulder trauma or surgery, Ahmed complain of stiffness and restricted active and passive movement have significantly worsened in all shoulder direction (Stiffness is most prominent) with substitute motion at scapula, mild Pain (3/10), mostly at the end of movement range, onset or duration of pain at 4- 12 months, with painful shoulder (now improved), the ADL activities or function severe difficulty with reaching overhead or behind, putting on a jacket, driving or working with tools, no swelling or deformity, no signs of impingement or rotator cuff tear, no tenderness, Joint capsule fibrosis and thickening have developed with Arthroscopy, No signs of joint effusion or acute inflammation



History

1-personal history

Name :Ahmed Elsayed

Age: 59

Sex :Male

occupation: mechanic.



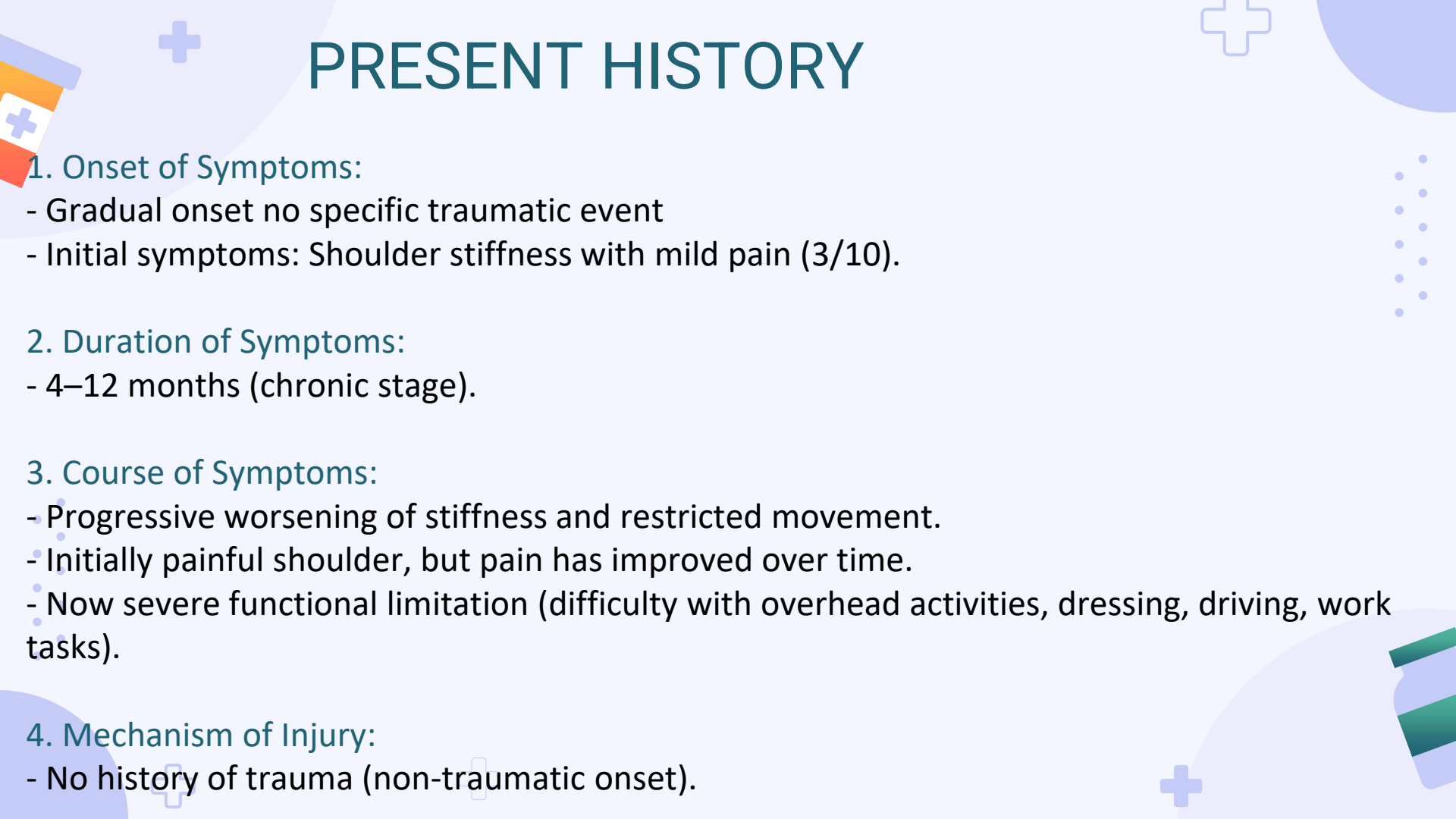
Diagnose

Adhesive Capsulitis (Frozen Shoulder) – Stage 2 (Fibrotic Phase)

Chief Complain

Severe shoulder stiffness limiting daily activities (overhead reach, dressing, driving, work)





PRESENT HISTORY

1. Onset of Symptoms:

- Gradual onset no specific traumatic event
- Initial symptoms: Shoulder stiffness with mild pain (3/10).

2. Duration of Symptoms:

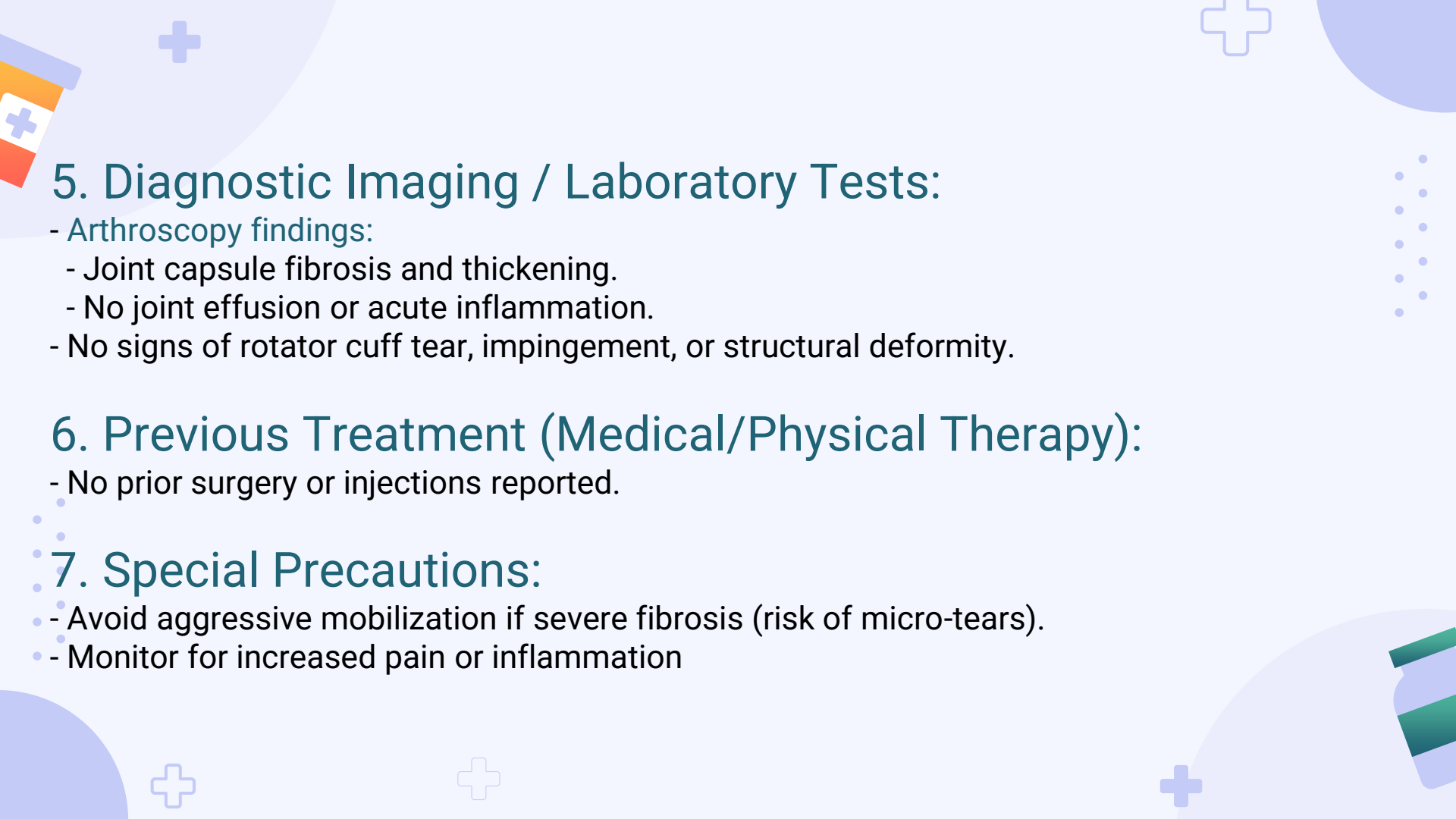
- 4–12 months (chronic stage).

3. Course of Symptoms:

- Progressive worsening of stiffness and restricted movement.
- Initially painful shoulder, but pain has improved over time.
- Now severe functional limitation (difficulty with overhead activities, dressing, driving, work tasks).

4. Mechanism of Injury:

- No history of trauma (non-traumatic onset).



5. Diagnostic Imaging / Laboratory Tests:

- Arthroscopy findings:
 - Joint capsule fibrosis and thickening.
 - No joint effusion or acute inflammation.
- No signs of rotator cuff tear, impingement, or structural deformity.

6. Previous Treatment (Medical/Physical Therapy):

- No prior surgery or injections reported.

7. Special Precautions:

- Avoid aggressive mobilization if severe fibrosis (risk of micro-tears).
- Monitor for increased pain or inflammation

PAIN HISTORY

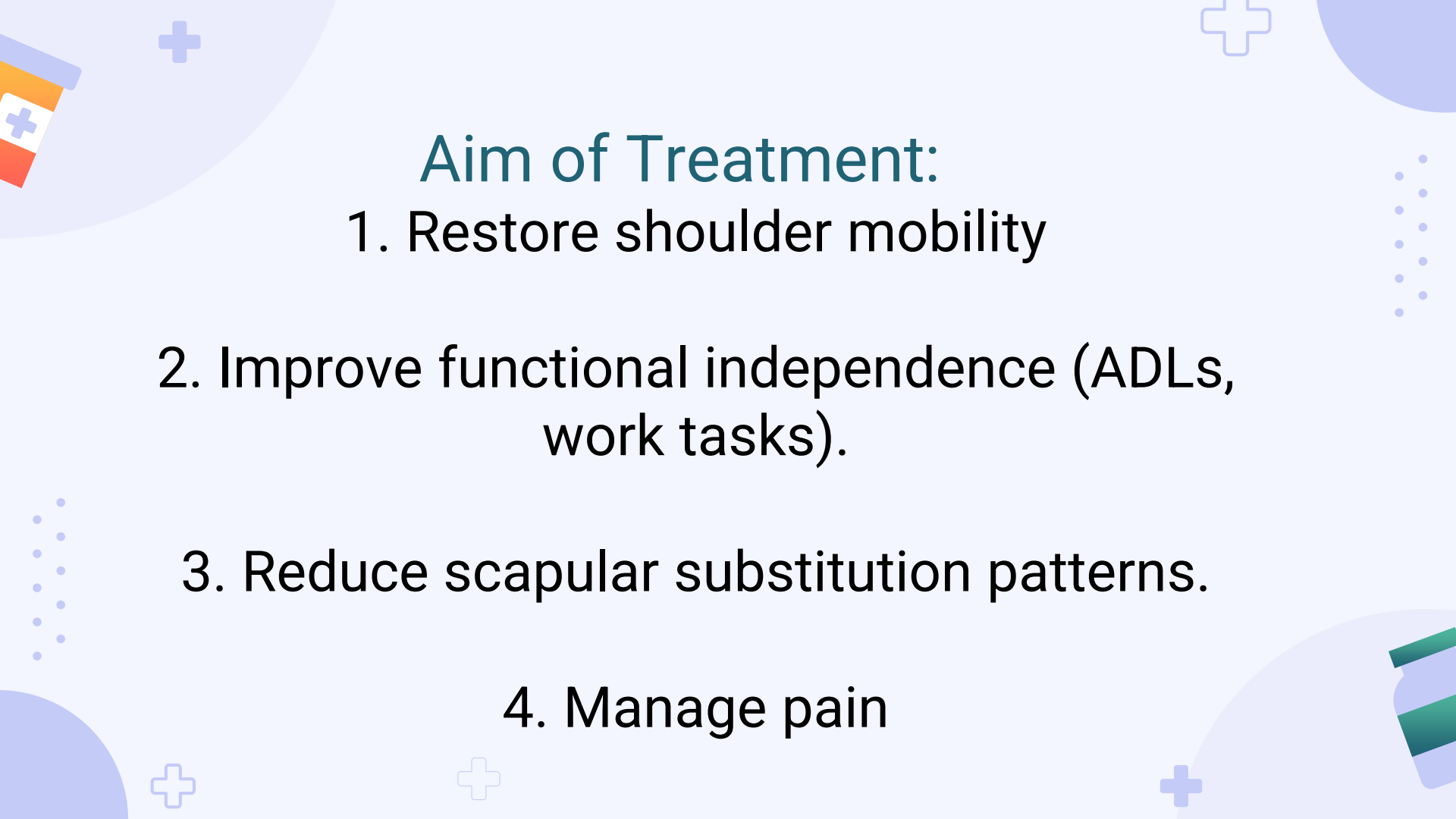
- 1. Quality: Pulling (at end-range movements)
- . Aggravating Factors:
 - Extreme shoulder movements (reaching overhead, behind the back).
 - Functional activities (putting on a jacket, driving, using tools)
- Mild, intermittent dull ache (3/10)

Assessment:

- Stage 2 Adhesive Capsulitis (fibrotic phase with capsular thickening).
- Severe functional impairment (ADLs and work-related tasks affected).
- No signs of rotator cuff pathology, impingement, or inflammation.

Problem List:

1. Severe restricted shoulder ROM (global limitation in flexion, abduction, IR/ER).
2. Scapulothoracic compensatory movements (due to glenohumeral stiffness).
3. Mild end-range pain (3/10).
4. Functional deficits (overhead reach, dressing, driving, tool use)
5. Capsular fibrosis



Aim of Treatment:

1. Restore shoulder mobility
2. Improve functional independence (ADLs, work tasks).
3. Reduce scapular substitution patterns.
4. Manage pain



Plan of Treatment:

- 1 Aggressive stretching (capsular-specific techniques):
 - 2 Posterior capsule stretch (cross-body adduction, sleeper stretch)
 - 3 Inferior capsule stretch (elevation with scapular stabilization).
External rotation (ER) / Internal rotation (IR)
 - 4 mobilization
Joint mobilizations (Grade III/IV for end-range capsular restriction)
 - 5 Scapular stabilization exercises
 - 6 Progressive AAROM/ROM exercises (wand, pulley, wall climbs).
- Home exercise program (consistent daily stretching).
- 



2. Pain Management

Heat therapy improve tissue elasticity

NSAIDs only if pain interferes with rehab

Non Physiotherapy management

3. Advanced Interventions (if conservative treatment fails):

- Hydrodilatation
- Manipulation Under Anesthesia
- Arthroscopic capsular release



4. Functional & Occupational Reintegration:

Task-specific training

Ergonomic modifications





*Role of regulatory authorities:

- Important role in preventing medication errors .
- Review of drug labels & nomenclature greatly enhances preventive strategies.
- FDA provides guidance to industry to maintain proper drug labels .
- FDA reviews 300 brand names in a year before they are marketed to avoid confusion of LASA drugs .
- Potentially confusing names rejected .
- FDA encourages pharmacists & other health professional to report any medication errors .